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Paleomagnetism of the Middle Proterozoic Laramie anorthosite complex and Sherman Granite, southern Laramie Range, Wyoming and Colorado

Steve S. Harlan and Lawrence W. Snee

U.S. Geological Survey, Denver, Colorado

John W. Geissman and Adrian J. Brearley

Department of Earth and Planetary Sciences, University of New Mexico, Albuquerque

Abstract. We present the results of a combined paleomagnetic and $^{40}Ar/^{39}Ar$ geochronologic investigation of the Middle Proterozoic Laramie anorthosite complex and Sherman Granite in the southern Laramie Range of Wyoming and Colorado. Anorthosites and monzosyenites of the Laramie anorthosite complex yield a well-defined characteristic magnetization of northeast declination (D) and moderate negative inclination (I), although antipodal normal and reverse polarity magnetizations are present at three sites. A grand mean direction from 29 of 35 sites in the complex is $D = 44.6°$, $I = -48.7°$ ($k = 77.4$, $\alpha_{95} = 3.1°$). Alternating field (AF) and thermal demagnetization behavior and rock magnetic experiments indicate that magnetization is carried by low-Ti titanomagnetite of single or pseudo-single domain character that occurs as elongate to rod-shaped inclusions in plagioclase and potassium feldspar. The Sherman Granite contains a dual polarity magnetization that is less well defined than that of the Laramie anorthosite complex but similar in declination and inclination ($D = 53.1°$, $I = -48.1°$, $k = 46.5$, $\alpha_{95} = 7.6°$, $n = 8/14$ sites); rock magnetic data indicate the primary carrier of remanence in Sherman Granite is magnetite. The $^{40}Ar/^{39}Ar$ geochronologic data from Sherman Granite hornblende, biotite, and microcline indicate that subsolidus cooling was moderate to relatively rapid through the range of temperatures over which magnetization was blocked and that the age of remanence is about 1415 Ma. Microcline data indicate that the Laramie anorthosite complex and Sherman Granite have probably not been thermally remagnetized. Paleomagnetic poles from the Laramie anorthosite complex and Sherman Granite are indistinguishable at the 95% confidence level, and individual virtual geomagnetic poles (VGPs) from both units are combined to provide a mean pole at 215.0°E, 6.7°S ($K = 46.9$, $A_{95} = 3.5°$, $N = 37$ VGPs). The location of this pole is similar to paleomagnetic poles derived from 1480 Ma to 1450 Ma intrusions elsewhere in North America, but it plots significantly north of those from Middle Proterozoic sedimentary strata of the Belt Supergroup and Sibley Group. In addition to the characteristic magnetization, samples from some sites in Sherman Granite contain a remanence of southeast declination and shallow negative inclination ($D = 154.9°$, $I = -16.0°$, $k = 90.6$, $\alpha_{95} = 9.7°$, $n = 4$ sites). This secondary magnetization was probably acquired during late Paleozoic time.

Introduction

Paleomagnetic studies of Precambrian rocks typically address one or more of the following questions. (1) What was the nature of the paleomagnetic field during the Precambrian and how long has the geodynamo existed? Related problems include: how do field reversal frequency, field intensity, and secular variation compare with that of the Phanerozoic? (2) Can reliable apparent polar wander (APW) paths be constructed for Precambrian cratons? (3) Can paleomagnetic data demonstrate that plate tectonics occurred during the Archean and Proterozoic, and if so, can they quantify relative motions between Precambrian cratons? Resolution of many of the preceding questions has been difficult owing to at least an order of magnitude less paleomagnetic data than for the Phanerozoic [*Roy*, 1983; *McWilliams*, 1981; *Van der Voo*, 1989] and the commonly poorly dated and discontinuous rock record of most Precambrian cratons. Correct interpretation of Precambrian paleomagnetic data is complicated by the great antiquity of many of the rock units studied, by the large analytical uncertainties associated with some isotopic dates, by the failure or inability to recognize and/or correct for structural deformation subsequent to remanence acquisition, and by the commonly complex overprint magnetizations caused by thermal, thermochemical, or chemical means. Consequently, varying interpretations regarding the viability or style of plate tectonics, based in part on selective interpretations of complex and commonly contradictory paleomagnetic data, have been proposed [e.g., *Piper*, 1976; 1982; *McElhinny and McWilliams*, 1977; *Irving and McGlynn*, 1981; *Burke et al.*, 1976a, b].

The style and extent of Precambrian tectonics, the construction of reliable Precambrian APW paths, and speculations concerning the existence and behavior of the ancient geomagnetic field can be elucidated only by using high-quality paleomagnetic data in which remanent magnetizations can be assigned geologically meaningful ages. Recent advances in high-precision isotopic dating techniques, modern methods of paleomagnetic data collection and analysis, and insights into the complex nature of thermochemical processes and remanent magnetization acquisition all have improved the quality of the Precambrian paleomagnetic database. In this paper,

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we present paleomagnetic data from two Middle Proterozoic plutons exposed in southern Wyoming and northern Colorado: the Laramie anorthosite complex and Sherman Granite. In addition, we report $^{40}Ar/^{39}Ar$ results from the Sherman Granite that provide information on the cooling history and age of remanence of both intrusions. Finally, we discuss the Laramie anorthosite complex/Sherman Granite pole in the context of a Middle Proterozoic APW path for North America.

Geologic Setting

The Laramie Range (Figure 1) is a narrow, north trending, basement-cored anticlinal uplift in southeastern Wyoming and north central Colorado that has been elevated along west dipping, low- to moderate-angle thrust faults that formed in response to horizontal crustal shortening during the Laramide Orogeny [*Brewer et al.*, 1982; *Tweto*, 1975]. The Laramie Range is bisected by the northeast projection of the Cheyenne belt, a zone of intense deformation that separates Archean (>2.5 Ga) crust of the Wyoming province to the north from Early Proterozoic crust with isotopic ages of 1.8 to 1.6 Ga to the south [*Karlstrom and Houston*, 1984; *Duebendorfer and Houston*, 1986, 1987; *Houston et al.*, 1989; *Geist et al.*, 1989]. The Cheyenne belt was interpreted as a suture produced during accretion of younger island arc terranes with the Archean Wyoming craton [*Duebendorfer and Houston*, 1986, 1987]. The exact location of the Cheyenne belt in the Laramie Range, however, is obscured by younger postkinematic plutons of the Middle Proterozoic Laramie anorthosite complex and Sherman Granite [*Geist et al.*, 1989].

Laramie Anorthosite Complex

The Laramie anorthosite complex (Figure 2) is a tabular to lens-shaped, composite, massif-type body that consists of anorthosite, mafic anorthosite (noritic anorthosite), troctolite, and associated hypersthene monzosyenite and hornblende monzosyenite plutons [*Fountain et al.*, 1981; *Geist et al.*, 1990; *Kolker and Lindsey*, 1989]. The complex is exposed over an area of 800 to 1000 km^2, and two distinct bodies have been identified. These are a northern domal or antiformal mass consisting primarily of anorthosite, mafic anorthosite, and subordinate mafic rocks, and a smaller southern body that consists of anorthosite [*Houston et al.*, 1989]. Seismic and gravity data indicate the complex is 4-5 km thick and may be floored by Sherman Granite [*Hodges et al.*, 1973; *Houston et al.*, 1989]. The Laramie anorthosite complex is cut by numerous northeast, northwest, and west-northwest trending shear zones and faults. The northern mass is dominated by a north-trending antiform defined by platy layering within the anorthosite and syenites; the structural culmination of the antiformal axis is marked by a prominent zone 300 to 900 m wide of crushed, granulated, and recrystallized anorthosite [*Newhouse and Hagner*, 1957]. *Fuhrman et al.* [1988] argue that deformation of the anorthosite mass occurred before the magma chamber completely solidified, giving rise to its domal structure. Paleomagnetic data from this study show that deformation of anorthosite occurred prior to acquisition of remanent magnetization during the Middle Proterozoic and is not due to Laramide folding as suggested by *Allmendinger et al.* [1982].

The anorthosite and mafic anorthosite consist primarily of plagioclase, with subordinate augite and hypersthene. A pronounced

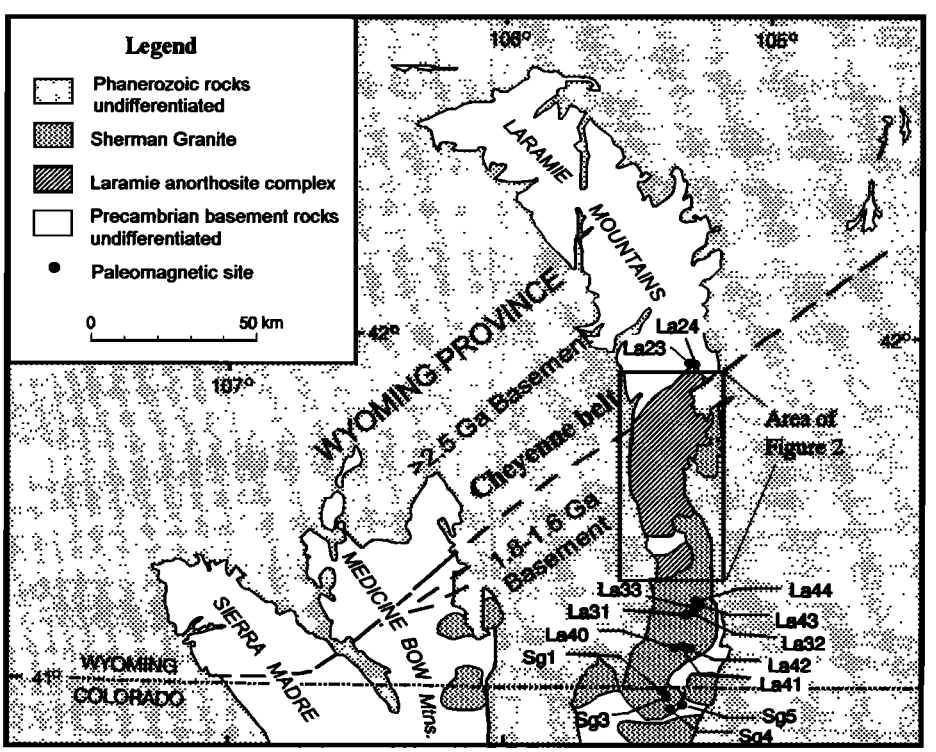


Figure 1. Generalized geologic map showing the distribution of Precambrian basement rocks in southeastern Wyoming and north central Colorado and the location of the Laramie anorthosite complex and Sherman Granite in the southern Laramie Range. The Cheyenne belt separates Archean basement rocks of the Wyoming province from Early Proterozoic basement to the south and east. Also shown are some of the paleomagnetic sites sampled in this study; additional sites are shown in Figures 2 and 3. Modified from *Hills and Armstrong* [1974].

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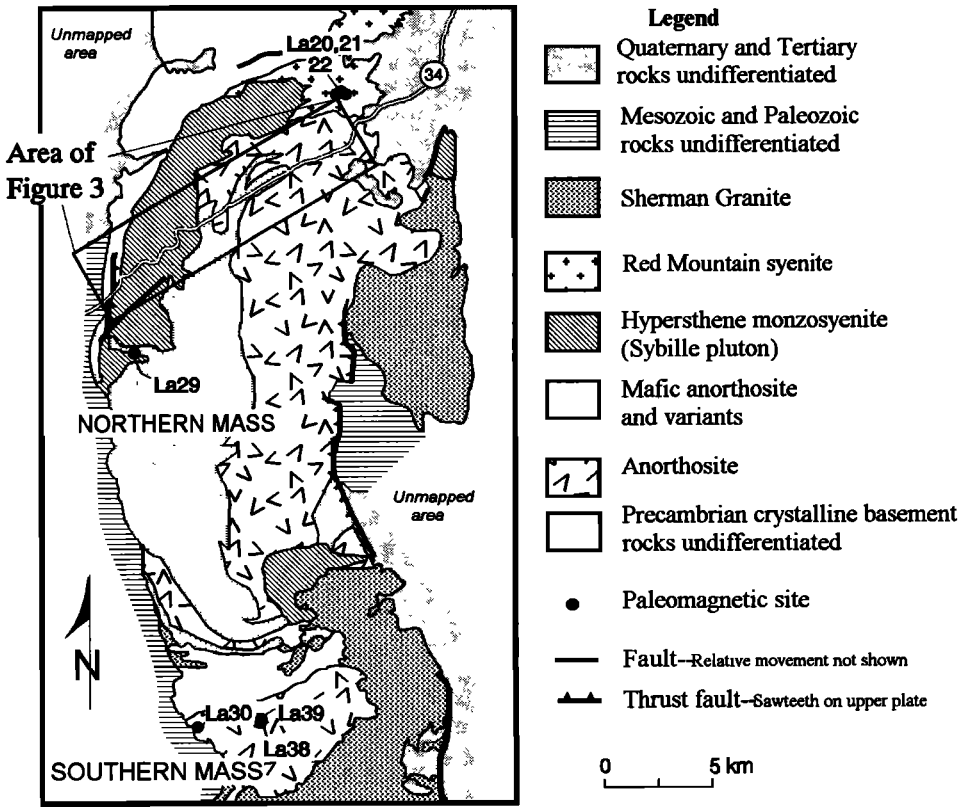


Figure 2. Geologic map of the Laramie anorthosite complex, southern Laramie Range, Wyoming. Modified from *Newhouse and Hagner* [1957].

cumulate layering is present within the anorthosite and, locally in the mafic anorthosite, large-scale rhythmic layering of felsic and mafic anorthosite has been reported [*Goldberg*, 1984]. The anorthosite was intruded by a hypersthene monzosyenite termed the Sybille pluton [*Fuhrman et al.*, 1988], the monzonitic Maloin Ranch pluton [*Kolker and Lindsey*, 1989; *Kolker et al.*, 1990], and hornblende monzosyenite and granite of the Red Mountain pluton [*Frost et al.*, 1990]. The Sybille monzosyenite consists of plagioclase, potassium feldspar (mesoperthite), and minor quartz, fayalitic olivine, clinopyroxene, and hypersthene [*Fountain et al.*, 1981]. The hornblende syenite of the Red Mountain pluton, located in the northern part of the complex, is composed of hornblende, microperthite, and plagioclase, with subordinate quartz [*Fountain et al.*, 1981; *Klugman*, 1966].

Field relationships, including the presence of anorthosite xenoliths within the hypersthene monzosyenite, indicate that the anorthosite and mafic anorthosite were emplaced prior to intrusion of the hypersthene monzosyenite, which was followed by intrusion of the hornblende monzosyenite. All phases of the Laramie anorthosite complex are cut by dikes, stringers, and plugs of Sherman Granite.

Sherman Granite

The Sherman Granite is a major composite pluton of batholithic dimensions exposed in the central and southern Laramie Range and Medicine Bow and Sierra Madre Mountains (Figure 1). Along the southern margin of the batholith, the 15-km-diameter Virginia Dale ring dike structure at the Wyoming-Colorado border may represent a deep-level exposure of a Middle Proterozoic caldera [*Eggler*, 1968]. The Sherman Granite is transitional between true granite and quartz monzonite and is petrologically and compositionally similar to Middle Proterozoic granites exposed elsewhere in Colorado [*Tweto*, 1987; *Geist et al.*, 1989]. Sherman Granite from the southern Laramie Range is pink to red, coarse-grained, equigranular to porphyritic granite that contains perthitic microcline, quartz, sodic plagioclase, biotite, and hornblende [*Geist et al.*, 1989]. Locally, microcline is mantled by plagioclase, resulting in a distinctive rapakivi texture [*Zielinski et al.*, 1981]. Accessory phases include ilmenite, magnetite, apatite, sphene, and zircon [*Geist et al.*, 1989]. In the ring dike complex, composite zones of quartz monzonite and diorite and small bodies of andesite and dacite are also present [*Eggler*, 1968].

The Sherman Granite was intruded by northwest trending mafic dikes, and both the Sherman Granite and dikes are cut by the Log Cabin batholith. The Laramie anorthosite complex, Sherman Granite, mafic dikes, and Log Cabin batholith are part of the Berthoud Plutonic Suite in Colorado and are related to a major belt of Middle Proterozoic magmatism that extended 4800 km from Labrador southwestward across the North American continent to California [*Anderson*, 1983].

Age of the Laramie Anorthosite Complex and Sherman Granite

The Laramie anorthosite complex and Sherman Granite have been subject to numerous geochronologic investigations, the most pertinent of which are summarized in Table 1. The anorthosites have not been directly dated, but their age is constrained by dates from the closely associated and perhaps comagmatic or cogenetic

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Table 1. Summary of Isotopic Dates From the Laramie Anorthosite Complex and Sherman Granite, Laramie Range, Southern Wyoming and Northern Colorado

	Rock Unit	Method	Age, Ma	Reference
Laramie anorthosite complex				
1	Monzonitic rocks, Sybille pluton	Pb-Pb zir	1432 ± 15, 1424 ± 15	*Subbarayudu* [1975]
2	Monzonitic rocks, Sybille and Maloin Ranch pluton	U-Pb zir	1435 ± 4	*Kolker et al.* [1991]
3	Monzonitic rocks, Red Mountain pluton	U-Pb zir	1439 +7, -6	*Frost et al.* [1990]
4	Monzonitic rocks, Red Mountain pluton	U-Pb zir	1440 ± 6	*Frost et al.* [1988]
5	Laramie anorthosite*	K-Ar hbl	1502 ± 30, 1525 ± 30	*Hills and Armstrong* [1974]
6	Laramie anorthosite*	K-Ar bio	1402 ± 28, 1411 ± 28	*Hills and Armstrong* [1974]
Sherman Granite				
1	Sherman Granite	Rb-Sr wr	1410 ± 30	*Peterman et al.* [1968]
2	Sherman Granite	Rb-Sr wr	1430 ± 20	*Zielinski et al.* [1981]
		Pb-Pb wr	1417 ± 100	
3	Sherman Granite	U-Pb zir	1430 ± 20	*Subbarayudu* et al. [1975]
4	Sherman Granite	U-Pb zir	1431 ± 6, 1412 ± 13	*Aleinikoff* [1983]
5	Sherman Granite**	K-Ar bio	1370 ± 50	*Ferris and Krueger* [1964]
6	Sherman Granite*	K-Ar bio	1376 ± 27	*Hills and Armstrong* [1974]
7	Sherman Granite*	K-Ar bio	1350 ± 27	*Hills and Armstrong* [1974]
8	Sherman Granite*	K-Ar hbl	1373 ± 27	*Hills and Armstrong* [1974]
9	Mafic dikes that cut the Sherman Granite**	K-Ar bio	1412 ± 70	*Ferris and Krueger* [1964]

Abbreviations are U-Pb, uranium-lead; Pb-Pb, lead-lead; Rb-Sr, rubidium-strontium; K-Ar, potassium-argon; zir, zircon; wr, whole rock; hbl, hornblende; and bio, biotite.

*Ages recalculated using the critical tables of *Dalrymple* [1992].

**Ages were recalculated from original data reported by *Ferris and Krueger* [1964] using the reviseddecay constants of *Steiger and Jäger* [1977].

Sybille, Maloin Ranch, and Red Mountain plutons. *Subbarayudu* [1975] reported $^{207}Pb/^{206}Pb$ zircon ages of 1432 ± 15 Ma and 1415 ± 15 Ma from monzonitic rocks of the Laramie anorthosite complex. Zircons from the Sybille and Maloin Ranch plutons yield U-Pb results that are only slightly discordant and give an upper intercept age of 1435 ± 4 Ma (unpublished data of R. E. Zartman reported by *Kolker et al.* [1991]). Zircons from the Red Mountain pluton, the youngest member of the Laramie anorthosite complex, give single crystal U-Pb zircon ages of 1440 ± 6 Ma [*Frost et al.*, 1988] and 1439 +7, -6 Ma [*Frost et al.*, 1990].

Whole rock Rb-Sr dates of 1410 ± 30 Ma and 1430 ± 20 Ma are reported from Sherman Granite [*Peterman et al.*, 1968; *Zielinski et al.*, 1981], whereas U-Pb and $^{207}Pb/^{206}Pb$ dates range from 1412 ± 13 to 1431 ± 6 Ma [*Aleinikoff*, 1983]. A lower age for Sherman Granite in Colorado is defined by the cross cutting Log Cabin batholith, which is dated at 1420 Ma by concordant whole rock and muscovite-microcline Rb-Sr isochrons [*Peterman et al.*, 1968; *Peterman and Hedge*, 1968].

In total, the Laramie anorthosite complex and Sherman Granite yield essentially identical isotopic dates that indicate emplacement of both plutons occurred between 1440 and 1430 Ma. The well-defined Rb-Sr isochron age of 1430 Ma reported by *Zielinski et al.* [1981] indicates that little, if any, disturbance of the Sherman Granite Rb-Sr system has occurred since emplacement. Complexities in U-Pb data, however, led *Aleinikoff* [1983] to conclude that the low-temperature history of the Sherman Granite may be complicated; a three-stage model was proposed in which discordance in the U-Pb system was attributed to Middle Proterozoic lead inheritance, crystallization, and late Paleozoic or Laramide lead loss [*Aleinikoff*, 1983]. The $^{40}Ar/^{39}Ar$ geochronologic data from this study are consistent with a relatively simple post emplacement history.

Previous Paleomagnetic Investigations

One of the earliest paleomagnetic investigations of Precambrian rocks of the western United States was conducted by *Eggler and Larson* [1968] as part of a study of the Virginia Dale ring-dike complex [*Eggler*, 1968]. Samples were collected from five sites in different rock types including "red granite" and quartz monzonite facies of Sherman Granite, mafic inclusions in Sherman Granite, and an andesite dike that cuts the granite. Samples were subjected to peak alternating field (AF) demagnetization of 35 to 40 mT, but none were thermally demagnetized. A well-defined dual polarity magnetization was obtained. Because most Pennsylvanian sedimentary rocks in the Virginia Dale area are nearly horizontal, tilt corrections were not applied to their data. The reported in situ mean direction is declination (D) = 52.0°, inclination (I) = -44.0° (k = 86.4, α_{95} = 4.3°, n = 14 samples from five sites) with a virtual paleomagnetic pole (VGP) at 8.0°S, 209.0°E (δp = 3.6°, δm = 5.5°). Stable remanence in the red granite facies was attributed to high-temperature oxidation of magnetite to hematite and finely disseminated hematite pigment along fractures and as inclusions in feld-

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spars; magnetite was interpreted to carry stable remanence in mafic inclusions [*Eggler and Larson*, 1968]. No rock magnetic data were presented in support of these interpretations.

Larson et al. [1973] reported a VGP from seven samples from one site in the Laramie anorthosite complex. The samples were found to be "stable" when subjected to demagnetization at 10 mT, or 400°C and 500°C. A VGP at 12°S, 201°E was reported, and this VGP is one of the most commonly cited Proterozoic poles from western North America. Calculation of a VGP from their in situ mean direction (after inversion to normal polarity), however, results in a VGP at 29.3°S, 246.0°E (δp = 3.3°, δm = 5.7°, 10 mT demagnetization). The 1500 Ma K-Ar age for the Laramie anorthosite reported by *Hills and Armstrong* [1974] and commonly assigned to this VGP [*Larson et al.*, 1973; *Piper*, 1988] is incorrect.

Finally, paleomagnetic data from the Laramie anorthosite complex and Sherman Granite were reported by *Kellogg* [1973] as part of an investigation of Proterozoic mafic dikes exposed in the Front Range of Colorado, northwest of Loveland. In his appendix, *Kellogg* [1973] reported site mean directions (5-8 samples/site) from Sherman Granite, an andesite dike that cuts the granite, Laramie anorthosite, and hypersthene syenite that is probably part of the Sybille monzosyenite. Samples were AF and thermally demagnetized, and data from individual sites averaged at the demagnetization interval that minimized dispersion. Not calculated by *Kellogg* [1973], these data give a mean direction of D = 50.7°, I = -46.4° (k = 35.6, α_{95} = 15.6°, n = 23 samples from four sites) and a VGP at 5.7°S, 211.1°E (K = 29.1, A_{95} = 17.3°).

Methods

Paleomagnetic samples were collected from 49 sites in the Laramie anorthosite complex and Sherman Granite (Figures 1, 2, and 3), and from two sites in adjacent Archean (?) basement rocks. The basement rocks did not yield usable paleomagnetic results and are not considered further. Most sites in anorthosite and syenitic rocks were located in roadcuts adjacent to Wyoming Highway 34 (Figure 3) where fresh, well-exposed material was available. Five sites were located in rocks exposed in two separate quarries, and two sites were from natural exposures. Samples of Sherman Granite were obtained from natural exposures of intrusions that cut the Laramie anorthosite complex, from roadcuts in the southern part of the range, from roadcuts along U.S. Interstate 80, or from roadcuts along U.S. Highway 287 in northern Colorado (Figure 1).

From 5 to 15 independently oriented samples were collected at each site using a portable drill equipped with a nonmagnetic drill bit, and samples were oriented using magnetic and solar compasses and a clinometer. Samples from most sites were collected over an area 2 to 5 m^2 in order to obtain a representative recording of the geomagnetic field and to average secular variation within site as fully as possible. The natural remanent magnetization (NRM) of each sample was measured using either cryogenic or spinner magnetometers. Detailed progressive AF and thermal demagnetization was conducted on two or more pilot samples from each site; the remaining samples were AF demagnetized in at least six steps up to a maximum field of 120 mT. Demagnetization results from each specimen were examined using orthogonal vector diagrams and stereographic projections. Characteristic directions, defined by three or more colinear points on vector diagrams, were calculated using a least squares method based on principal component analysis (PCA) [*Kirschvink*, 1980]. Site and group mean directions were calculated using *Fisher* [1953] statistics.

Magnetic carriers were investigated using a combination of rock magnetic and electron beam techniques. Rock magnetic methods included acquisition of isothermal remanent magnetization (IRM), comparison of AF demagnetization of IRM and NRM, and Curie-temperature determinations. Observation and characterization of magnetic phases from the Laramie anorthosite complex were made using scanning transmission electron microscopy (STEM). For STEM analyses, selected feldspar crystals from monzosyenites and anorthosites were selected from doubly polished sections and pre-

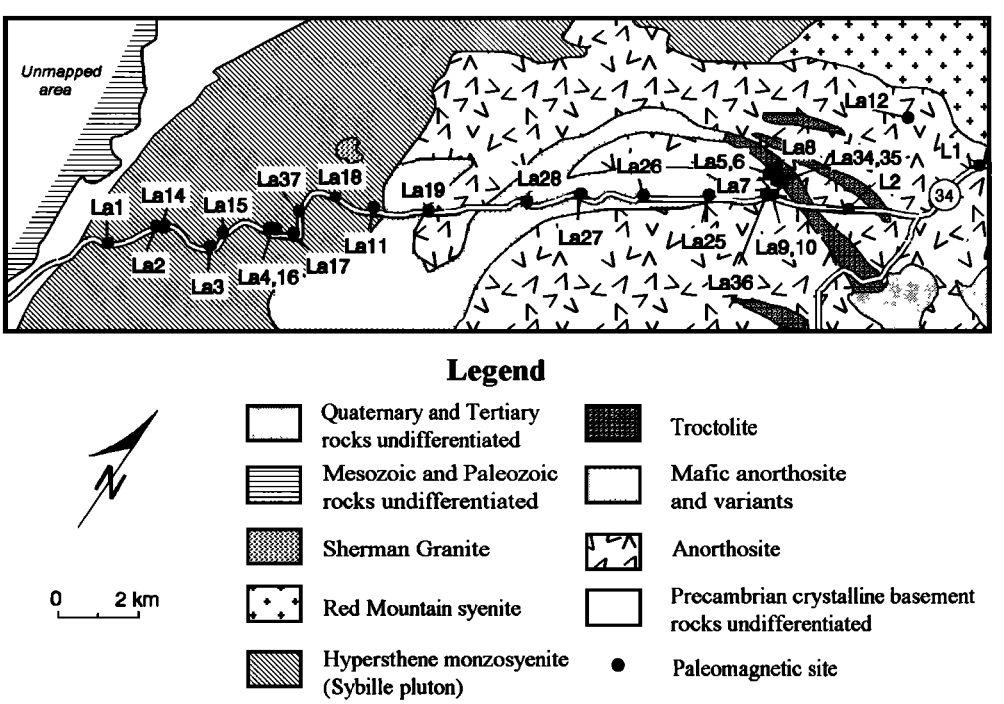


Figure 3. Geologic map of the Laramie anorthosite complex showing paleomagnetic sites collected along and near Wyoming State Highway 34, southern Laramie Range, Wyoming (modified from *Newhouse and Hagner* [1957]).

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pared by ion beam bombardment [*Barber*, 1970]. All samples were analyzed using a JEOL 2000FX analytical microscope at 200 kV accelerating voltage. Quantitative analytical data were obtained in situ by energy dispersive analysis (EDS). Magnetic phases were identified using a combination of X ray microanalysis, conventional electron diffraction, and microdiffraction [e.g., *Brearley and Champness*, 1986].

Samples for $^{40}Ar/^{39}Ar$ dating were prepared from two samples of Sherman Granite. The samples were crushed and sieved, and mineral separates were produced by standard heavy liquid techniques and hand-picking. Mineral separates and neutron flux monitors were wrapped in aluminum foil packages, placed in quartz vials, and irradiated for 96 or 100 hours in the U.S. Geological Survey TRIGA reactor. Variations in neutron fluence within the irradiation package were monitored by 6 to 10 standards along the length of each vial; the monitor used was hornblende MMHb-1, with an assigned K-Ar age of 520.4 Ma [*Samson and Alexander*, 1987; *Alexander et al.*, 1978; *Dalrymple et al.*, 1981]. Corrections for interfering isotopes were made by measuring the argon isotopes of K_2SO_4 and CaF_2 irradiated in each package. Each sample was heated in a double-vacuum resistance furnace for 20 min in a series of 10 to 20 steps to a maximum temperature of 1450°C. Five naturally occurring and radiogenic isotopes of argon (^{40}Ar, ^{39}Ar, ^{38}Ar, ^{37}Ar, and ^{36}Ar) were measured at each step using a mass spectrometer, and isotopic compositions were corrected for mass discrimination. Apparent ages were calculated using the decay constants recommended by *Steiger and Jäger* [1977]. Where appropriate, age plateaux were determined by inspection of contiguous gas fractions using the critical value test of *Dalrymple and Lanphere* [1969]. A detailed description of analytical procedures similar to those used in this study are given by *Tysdal et al.* [1990, Appendix 1].

Paleomagnetic Results

Laramie Anorthosite Complex

In situ NRM directions from the anorthosite and syenitic rocks from Laramie anorthosite complex show moderate dispersion about a northeast, moderate negative inclination mode (Figure 4a). No apparent streaking of NRM directions toward the present-day field or other direction is apparent. NRM intensities ranged from 0.003 A/m to 3.0 A/m and averaged 0.4 A/m. Approximately 27% of samples analyzed failed to yield meaningful results after demagnetization. Of these, 85% showed extreme erratic directional and intensity changes during AF or thermal demagnetization. This included all samples from syenite dikes, one site in hypersthene syenite of the Red Mountain pluton, and two sites in natural exposures of anorthosite in the southern anorthosite mass. The remaining 15% of samples gave coherent demagnetization trajectories after demagnetization, but the observed directions were grossly inconsistent with the majority of samples within site; these samples were rejected from calculation of site mean directions. The 73% of usable samples typically gave well-defined monotonic decay of the remanence vector to the origin of demagnetization diagrams and directions of northeast declination and moderate negative inclination (Figures 5a-5d); this direction is termed the characteristic magnetization of the Laramie anorthosite complex. Secondary magnetizations are inconsistent in direction and removed by AF

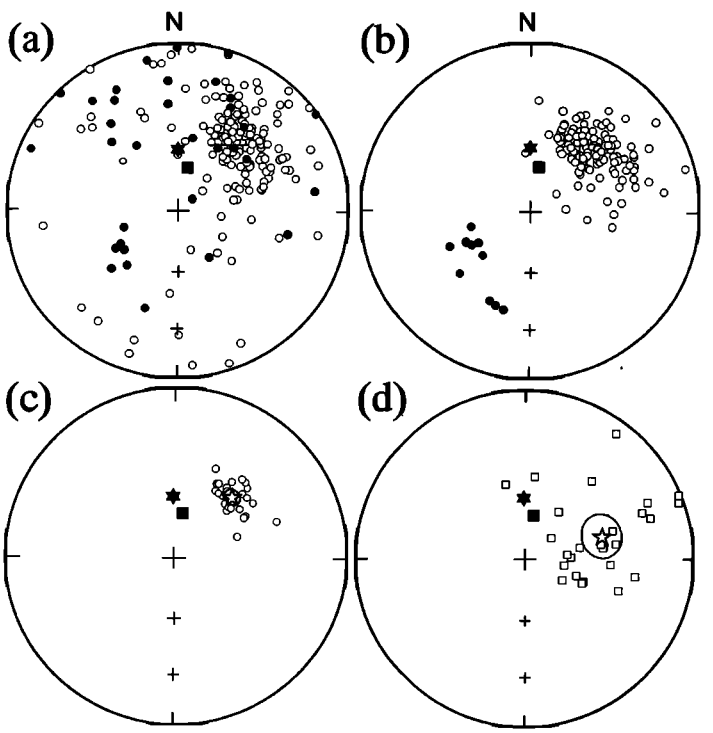


Figure 4. Equal area projections showing in situ paleomagnetic results from the Laramie anorthosite complex: (a) natural remanent magnetization directions, (b) characteristic directions used in the calculation of site mean directions, (c) In situ site mean directions (n = 29), grand mean (five-pointed star), and α_{95} cone of confidence, and (d) structurally corrected site means (n = 27) after tilt corrections based on the attitude of igneous layering, and its grand mean direction and α_{95}. Open circles are projections on the lower hemisphere, and solid circles are projections on the upper hemisphere. The directions of the time-averaged and present-day geomagnetic field directions are shown by a six-pointed star and square, respectively.

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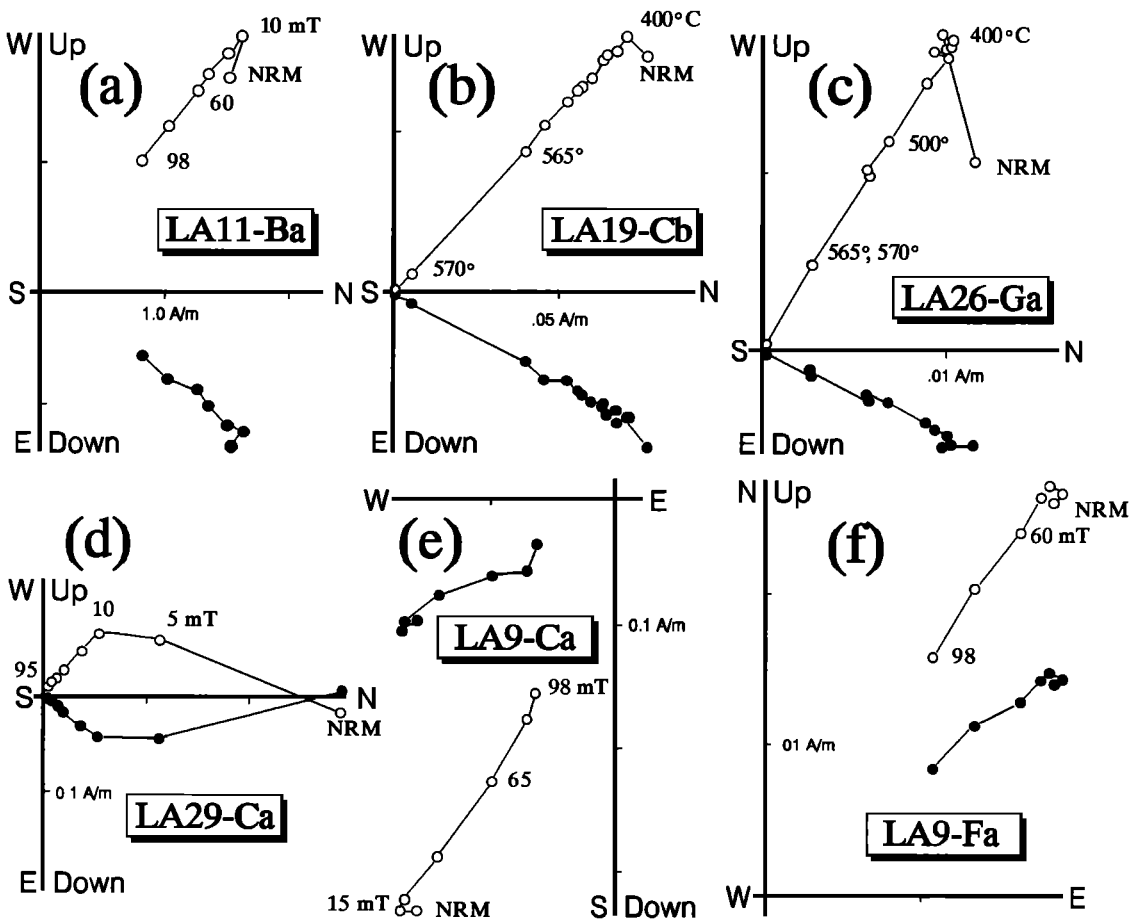


Figure 5. Orthogonal vector diagrams showing representative alternating field and thermal demagnetization behavior of anorthosite and syenite samples from sites in the Laramie anorthosite complex. Open and solid symbols represent the in situ magnetization components projected onto the vertical and horizontal planes, respectively. Demagnetization is in millitesla (mT) and degrees Celsius. Note that the coordinates of the projections in Figures 5e and 5f have been shifted in order to better illustrate the magnetizations of opposite polarity at site La9.

demagnetization of 5 to 10 mT or temperatures of 150° to 200°C. At three sites, samples with southwest declination and moderate positive inclination, antipodal to the characteristic direction, coexist with samples of negative inclination (Figures 5e and 5f). Most samples are characterized by moderate to high median destructive fields and by a narrow range of laboratory unblocking temperatures below 580°C (Figures 5b and 5c).

Sample directions determined using PCA from acceptable samples are well grouped (Figure 4b). Because much of the remanence remained at 100 mT (Figures 5a, 5d, 5e, and 5f), most least squares fits from AF demagnetized samples were anchored to the origin of their vector diagrams. The well-defined decay to the origin displayed in demagnetization diagrams by all thermally demagnetized samples (Figures 5b and c), indicating the presence of a single component throughout the demagnetization interval, demonstrates that such an approach is justified. Maximum angular deviation (MAD) values for linear segments anchored to the origin commonly ranged from 0.5° to 3.0°; nonanchored fits usually produced identical directions, but MAD values tended to be somewhat greater (i.e., 3.0° to 7.0°).

Site mean directions (Figure 4c) are generally well defined, with most individual site k values ranging from 25 to 475, and α_{95} values from 2.5° to 10.0°, although three sites show somewhat greater dispersion (Table 2). To investigate the origin of the antiformal structure of the northern domal mass, we conducted a fold test on 27 sites where the attitude of igneous layering could be determined. The in situ mean for the 27 sites used in this analysis is $D = 44.9°$, $I = -48.9°$ ($k = 75.9$, $\alpha_{95} = 3.2°$). Application of tilt corrections results in a profound divergence of site mean directions, and a structurally corrected mean of $D = 72.7°$, $I = -50.7°$ ($k = 8.5$, $\alpha_{95} = 10.2°$) (Figure 4d). The ratio of in situ to structurally corrected k values for a *McElhinny* [1964] fold test is 8.93, significantly greater than the 1.93 needed for significance at p = 0.01; the fold test is thus negative and significant at the 99% confidence level. Doming of the northern mass of the Laramie anorthosite complex occurred prior to acquisition of the characteristic magnetization. Calculation of a grand mean direction for the Laramie anorthosite complex based on all in situ site means gives a mean direction of $D = 44.6°$, $I = -48.7°$ ($k = 77.4$, $\alpha_{95} = 3.1°$, $n = 29/35$ sites) (Figure 4c; Table 3).

In IRM experiments, all anorthositic and syenitic samples achieved IRM saturation by 150 to 300 mT and gave coercivity of remanence (H_{cr}) values of 20 to 80 mT (Figure 6a-f) and median destructive fields of 20 to 100 mT. These data indicate the dominance of a low to moderate coercivity phase as the primary carrier of remanence. The shapes of the IRM curves (Figures 6a to f) are similar to those of single-domain or pseudo-single-domain magnetite [*Dunlop*, 1981]. Thermal demagnetization of remanent magnetization with narrow unblocking temperature spectra and

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Table 2. Site Mean Paleomagnetic Data From the Laramie Anorthosite Complex and Sherman Granite, Wyoming and Colorado

Site	Rock Type	Slat. °N	Slong. °E	n/N_0	*D*, deg	*I*, deg	Strike, deg	Dip, deg	SD, deg	SI, deg	*k*	α_{95}, deg	Plat. °N or °S	Plong. °E
						Laramie Anorthosite Complex								
L1	An	41.800	254.670	5/10	48.3	-55.8	110	50	35.6	-8.8	215	5.2	-0.3	217.6
L2	An	41.782	254.675	10/10	40.4	-39.0	0	50	345.0	-52.2	46	7.2	-16.1	215.9
L3	Sy	41.708	254.483	7/8	47.1	-53.6	240	23	78.4	-52.4	51	7.4	-2.7	217.2
La1	Sy	41.701	254.499	7/8	46.1	-40.4	--	--	--	--	93	6.3	-12.4	211.9
La2	Sy	41.709	245.511	11/11	43.4	-48.9	246	30	80.6	-51.2	174	3.5	-8.0	217.6
La3	Sy	41.709	254.525	7/7	44.1	-49.5	246	30	81.8	-51.2	229	4.0	-7.2	217.4
La4	Sy	41.718	254.538	6/6	36.8	-47.5	208	50	71.4	-23.2	116	6.3	-11.9	222.1
La5*	An	51.780	254.650	5/7	24.5	-41.5	280	35	51.2	-73.5	26	15.3	-20.6	230.7
La6	An	41.780	254.652	7/8	42.6	-39.2	280	35	77.6	-63.1	29	11.5	-14.9	214.2
La7	An	41.778	254.651	7/8	53.1	-50.8	280	35	108.7	-64.1	88	6.5	-1.9	211.6
La8	Tr	41.782	254.652	7/9	59.7	-50.0	280	35	111.4	-39.8	12	18.1	1.0	206.7
La9*	An	41.777	254.655	9/9	47.2	-54.2	278	25	87.6	-67.9	40	8.2	-2.1	217.5
La10**	An	41.777	254.654	0/8										
La11	Sy	41.735	254.563	9/9	54.5	-43.2	192	50	69.4	-3.6	55	7.0	-6.3	206.8
La14	Sy	41.709	254.515	9/9	30.6	-48.1	203	20	50.0	-42.0	202	3.6	-13.7	227.4
La15	An xen	41.713	254.525	5/7	71.8	-58.3	45	21	34.9	-61.5	51	9.5	13.8	205.1
La16	Sy	41.716	254.539	10/10	37.1	-51.9	28	25	6.2	-49.0	120	4.4	-8.3	223.7
La17	Sy	41.719	254.542	8/8	41.1	-48.4	235	35	79.8	-44.8	476	2.5	-9.4	219.1
La18	Sy	41.732	254.549	8/8	29.8	-48.4	206	50	68.2	-26.5	46	8.3	-13.7	228.1
La19	An	41.738	254.571	9/9	31.1	-51.3	195	50	64.7	-21.6	34	14.0	-10.9	228.1
La20**	Sy	41.882	254.736	0/11										
La21**	Sy dike	41.882	254.722	0/6										
La22**	Sy dike	41.903	254.715	0/9										
La25	An	41.770	254.639	9/10	70.4	-36.4	275	45	109.0	-41.0	147	4.3	-0.3	192.5
La26	An	41.764	254.626	7/8	38.2	-48.9	278	45	119.6	-69.2	58	8.0	-10.2	221.6
La27	An	41.758	254.610	7/9	48.3	-53.6	235	50	99.1	-34.8	37	10.0	-2.7	217.2
La28	An	41.751	254.598	8/8	50.1	-43.8	192	55	67.4	-1.8	48	8.0	-8.3	210.3
La29	Sy	41.645	254.515	9/9	37.0	-43.3	250	50	93.9	-48.1	194	3.7	-14.8	220.3
La30	An	41.392	254.559	8/8	34.3	-52.0	--	--	--	--	293	3.2	-9.2	225.9
La34	An	41.775	254.653	3/5	40.6	-50.4	280	35	98.7	-71.1	68	9.8	-8.1	220.4
La35	An	41.773	254.654	4/6	60.1	-50.9	280	35	113.2	-41.2	94	7.2	1.8	207.0
La36*	An	41.778	254.590	9/11	43.7	-54.3	278	25	83.8	-69.4	34	8.9	-3.6	220.0
La37	An	41.725	254.541	7/7	46.5	-44.3	240	25	71.7	-44.7	126	5.4	-9.8	213.2
La38**	An	41.399	254.613	0/11										
La39**	An	41.397	254.619	0/9										
						Sherman Granite								
La12*	ShGr			0/22										
La24	ShGr	41.849	254.658	9/9	84.6	-41.0	--	--	--	--	30	9.6	11.6	185.8
La31	ShGr	41.255	254.565	9/9	41.8	-50.5	--	--	--	--	77	5.9	-7.9	219.5
La32*	ShGr	41.255	254.566	0/12										
La33	ShGr	41.254	254.585	9/10	43.5	-48.3	--	--	--	--	25	10.5	-8.8	217.2
La40	Sy xen	41.235	254.563	5/8	61.8	-47.5	--	--	--	--	30	14.2	0.2	203.9
La41	ShGr	41.236	254.563	4/9	81.8	-45.7	--	--	--	--	16	23.8	11.8	190.4
La42	ShGr	41.232	254.563	9/10	52.1	-46.9	--	--	--	--	17	14.0	-5.5	210.2
La43	ShGr	41.253	254.585	2/9	34.7	-57.0	--	--	--	--	--	--	-5.0	227.7
La44	ShGr	41.254	254.565	8/11	60.4	-42.6	--	--	--	--	27	10.9	-3.6	202.3
Sg1	ShGr	40.991	254.590	5/13	53.8	-47.0	--	--	--	--	8	24.0	-4.8	209.0
Sg3	ShGr	40.960	254.640	4/8	215.0	44.2	--	--	--	--	26	18.5	-16.5	221.8
Sg4	ShGr	40.951	254.655	14/14	238.0	47.3	--	--	--	--	43	6.2	-2.5	206.2
Sg5	ShGr	40.901	254.723	1/23	87.3	-51.3	--	--	--	--	--	--	18.5	191.2
						Sherman Granite -- Late Paleozoic Overprint								
La41	ShGr	41.236	254.563	2/9	147.3	-15.0	--	--	--	--	--	--	45.7	124.7
La42	ShGr	41.232	254.563	1/9	156.0	-15.0	--	--	--	--	--	--	50.2	133.7
Sg1	ShGr	40.991	254.590	3/13	166.9	-18.1	--	--	--	--	20	28.4	56.3	98.4
Sg2	ShGr	40.992	254.590	14/23	149.7	-15.3	--	--	--	--	45	6.0	46.7	121.5

Rock types are An, anorthosite; Sy, monzosyenite and syenite; Sy dike, syenitic dikes in anorthosite; Am, amphibolite; Tr, troctolite; ShGr, Sherman Granite plus compositional variants and related rocks; An xen, anorthosite xenolith in monzosyenite; Sy xen, syenitic xenolith within the Sherman Granite. Slat. and Slong. = latitude and longitude of the sample collection site; n/N_0 is the number of samples used in the calculation of site mean direction/samples measured; *D* and *I* are declination and inclination of the in situ mean direction; SD and SI are declination and inclination of the site mean direction after tilt correction for the attitude of igneous layering; strike and dip corrections are from measurements of igneous layering made during this study or have been extrapolated from mapping by *Newhouse and Hagner* [1957] and the dip direction is given as 90° clockwise from the strike direction; *k* is the best estimate of the "precision" or concentration parameter [*Fisher*, 1953]; α_{95} is the semiangle of the cone of 95% confidence about the site mean; Plat. and Plong. are latitude and longitude of the virtual geomagnetic pole based on the in situ site mean direction and inversion to normal polarity. Positive values of Plat. plot in the northern hemisphere, negative values in the southern hemisphere.

*Sites that contain samples of normal and reverse polarity samples within the same site.

**Samples from this site were unstable during demagnetization and a characteristic dirction was not isolated.

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Table 3. Mean Paleomagnetic Directions and Paleomagnetic Poles From the Laramie Anorthosite Complex and Sherman Granite, Southern Laramie Range, Wyoming

	n/N	*D*, deg	*I*, deg	*k*	α_{95}, deg	δ_{63}, deg	Plat. °N or °S	Plong. °E	*K*	A_{95}, deg	δ_{63}, deg
Laramie anorthosite complex	29/35	44.6	-48.7	77.4	3.1	9.2	-7.4	216.8	57.4	3.6	10.7
Sherman Granite	12/14	58.3	-48.7	38.9	7.1	13.0	-1.1	207.2	24.3	9.0	16.5
	8/14*	53.1	-48.1	46.5	7.6	11.9	-4.3	210.5	30.7	9.4	14.6
Combined Laramie anorthosite complex and Sherman Granite	37/49	46.9	-48.4	65.9	2.9	10.0	-6.7	215.0	46.9	3.5	11.9
Sherman Granite -- Late Paleozoic overprint	$n = 4$	154.9	-16.0	90.6	9.7	8.5	50.1	115.5	86.5	9.9	8.7

Ratio *n/N* is number of sites used in calculation of the site mean direction or paleomagnetic pole versus the total number of sites measured: *D* and *I* are declination and inclination of the mean direction; *k* is the best estimate of the concentration parameter of the mean direction: α_{95} is semi-angle of 95% cone of confidence of the mean direction; δ_{63} is angular standard deviation of the group mean direction and the virtual geomagnetic pole: Plat. and Plong. are latitude and longitude of the paleomagnetic pole; *K* is precision parameter of the paleomagnetic pole; A_{95} is 95% cone of confidence of the paleomagnetic pole.
*Mean direction and pole calculated after rejecting sites with α_{95} greater than 20° or with only two samples of the characteristic direction.

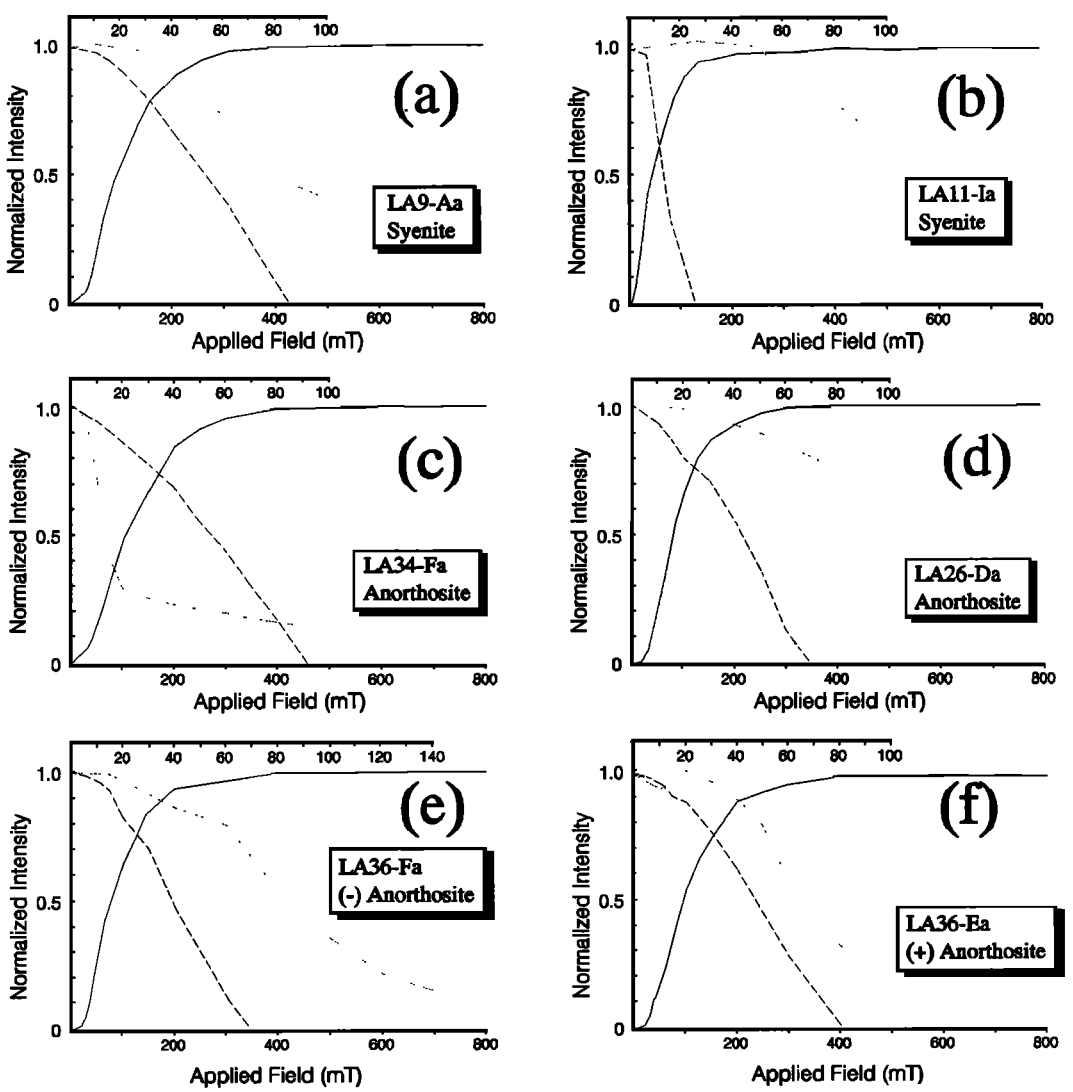


Figure 6. Normalized intensity diagrams of selected Laramie anorthosite complex samples showing the acquisition of isothermal remanent magnetization (IRM) (solid line; lower scale) caused by the application of an applied direct current field; alternating field (AF) demagnetization of natural remanent magnetization (dotted line; upper scale); and AF demagnetization of IRM (dashed line; upper scale). Plus and minus signs in Figures 6e and 6f refer to normal and reverse polarity samples from the same site.

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complete unblocking below 580°C (Figures 5b and c), as well as Curie temperature determinations of 560° to 580°C (not shown), are consistent with the presence of low-titanium titanomagnetite in anorthosites and monzosyenites of the Laramie anorthosite complex. None of the rock magnetic or thermal demagnetization experiments indicate the presence of hematite in the Laramie anorthosite complex samples.

STEM and paleomagnetic microanalysis experiments indicate that the remanence carriers in samples from the Laramie anorthosite complex lie within individual feldspar phenocrysts [*Geissman et al.*, 1988]. In particular, STEM analyses demonstrate the presence of abundant, generally euhedral, commonly elongate to acicular magnetic phases dispersed within individual feldspar grains. In plagioclase from anorthosite La28, such oxides are less than 0.3 μm wide and as great as 1.0 μm long (Figure 7a). X ray microanalysis and electron diffraction data indicate these grains are titanomagnetite and in some cases are zoned from pure magnetite to titanomagnetite (i.e., 20 wt% TiO_2) along the length of the crystal [*Geissman et al.*, 1988]. STEM investigation of a monzosyenite plagioclase crystal from site La17 revealed rare regions of acicular opaque rods, 1 to 2 μm in length, identified as magnetite. Alkali feldspars from the sample were found to contain titanomagnetite 0.1 to 0.2 μm long that occurred exclusively within lamellae where peristerite exsolution had occurred (Figure 7b). The size and aspect ratios of the magnetite grains and the rock magnetic properties exhibited by Laramie anorthosite complex samples are consistent with those of single-domain or pseudo-single-domain magnetite. Similar needle- or rod-shaped magnetites contained within feldspar grains have been reported to be the most probable carriers of stable remanence in other similar anorthosite bodies and in some mafic and intermediate composition intrusions [*Hargraves and Young*, 1969; *Murthy et al.*, 1971; *Wu et al.*, 1974; *Morgan and Smith*, 1981; *Davis*, 1981; and *Scofield and Roggenthen*, 1986].

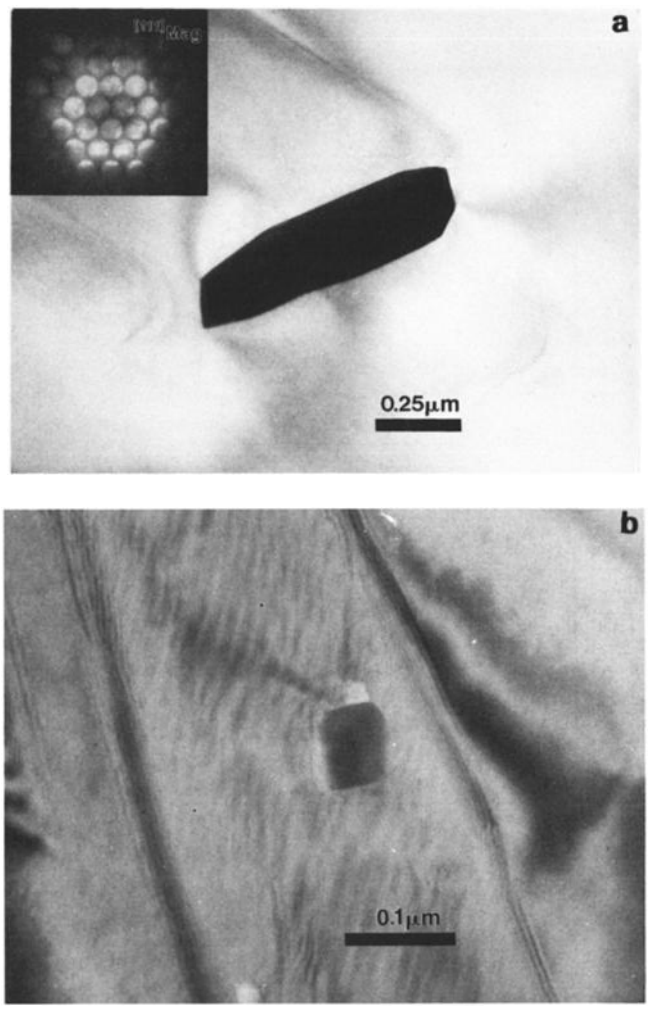


Figure 7. Scanning transmission electron microscope (STEM) images of submicron titanomagnetite contained as inclusions within plagioclase and potassium-feldspar phenocrysts in paleomagnetic samples from the Laramie anorthosite complex. (a) Elongate magnetite crystal in plagioclase from anorthosite site La28. The crystal is oriented with [111] parallel to the electron beam; inset shows convergent beam microdiffraction pattern. Energy dispersive (EDS) X ray spectra (not shown) reveal that magnetite and titanomagnetite-rich regions occur within individual magnetite crystals. (b) Magnetite crystal within peristerite lamellae in alkali feldspar from a monzosyenite from site La17; EDS X ray analysis indicates that this sample is nearly pure magnetite.

Sherman Granite

In situ NRM directions from Sherman Granite and related rocks are scattered but define a northeast, moderately negative inclination distribution similar to that of the Laramie anorthosite complex (Figure 8a). A tendency toward shallowly inclined NRM directions may be caused by a present-day viscous remanent magnetization. NRM intensities for granitic samples varied from 0.00026 A/m to 0.27 A/m, whereas intensities for diorites ranged from 0.1 A/m to 1.6 A/m. Unfortunately, a large number of Sherman Granite samples were characterized by poor demagnetization behavior, and only 62% of samples analyzed gave useful results. The rest either failed to yield stable endpoints or remagnetization circles, gave noisy demagnetization trajectories, or gave directions that appeared randomly oriented with respect to other samples within site. Of samples yielding acceptable results, 80% gave directions of northeast declination and moderate negative inclination (Figures 9a and 9b) or were approximately antipodal (Figure 9c), and 20% were of southeast declination and moderate negative inclination (Figures 9d and 9e). Because of its similarity to the Laramie anorthosite complex, the former are considered characteristic of the Sherman Granite, and the latter are suspected to be a secondary magnetization of much younger age.

Samples that contain the characteristic magnetization typically display low coercivities and distributed laboratory unblocking temperatures (Figures 9a and 9b). Secondary directions are usually of northerly declination and positive inclination and were removed by 5 to 10 mT or 200° to 400°C. The direction and coercivity/unblocking temperature characteristics indicate that these overprints are probably a present-day VRM. The characteristic magnetization is usually isolated in AF demagnetization of 5 to 30 mT for granite samples and 10 to 70 mT for diorites. Diorites are completely unblocked by 585°C, and usually less than 5% of the NRM in granites remained at this temperature. Magnetizations existing in granite samples above this temperature are either inconsistently oriented or erratic, possibly due to the creation of new magnetic phases at elevated temperature (e.g., Figure 9a). The characteristic direction for samples with MAD values less than 10° is of northeast declination and moderate negative inclination (Figure 8b). At two sites near the southern margin of the Sherman Granite, antipodal magnetizations were obtained.

Characteristic directions obtained from the Sherman Granite are similar to those of the Laramie anorthosite complex, although a tendency toward more easterly directions is apparent (Figure 8b).

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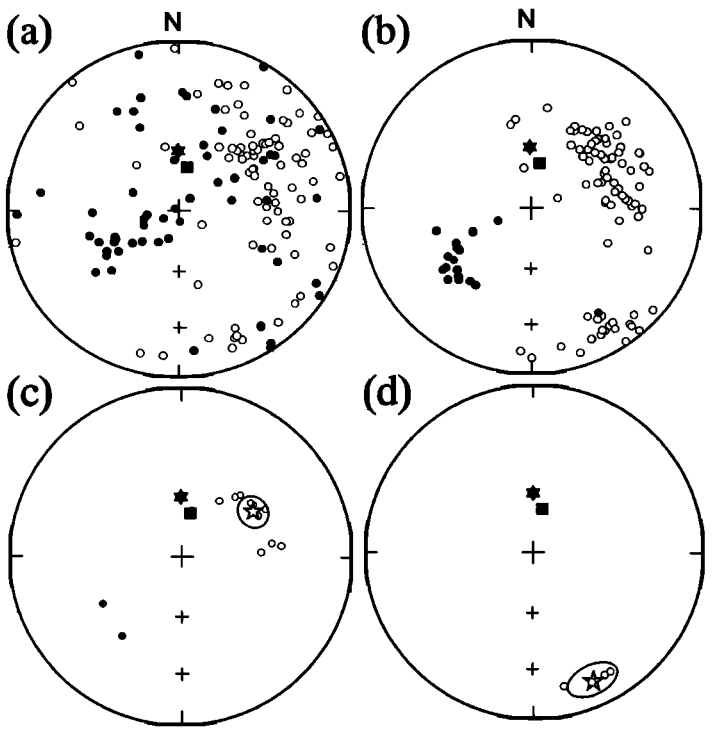


Figure 8. Equal area projections showing in situ paleomagnetic results from the Sherman Granite: (a) natural remanent magnetization directions, (b) characteristic directions used in the calculation of site mean directions, (c) site mean directions and grand mean and its α_{95} cone of confidence. The grand mean direction is based on sites with $n \geq 3$ and $\alpha_{95} < 20°$, and (d) late Paleozoic site and mean directions. Conventions are as in Figure 4.

Some of this streaking may be due to contamination of the characteristic direction by a secondary magnetization, described below. Dispersion within individual site mean directions is somewhat greater than most sites in the Laramie anorthosite complex (i.e., α_{95} ranged from 6.0° to 24.0° and k values from 8 to 77; Table 2). The Sherman Granite grand mean is $D = 58.3°$, $I = -48.7°$ ($k = 38.9$, $\alpha_{95} = 7.1°$) for 12 sites (Table 3). Excluding sites with large dispersion (i.e., $\alpha_{95} > 20°$) or site directions determined from only one or two samples, an essentially identical grand mean direction ($D = 53.1°$, $I = -48.1°$, $k = 46.5$, $\alpha_{95} = 7.6°$, $n = 8$ sites) is obtained (Figure 8c and Table 3).

In addition to the characteristic direction, southeast, shallow negative inclination magnetizations were found at four sites, including the regolith breccia at the Precambrian-Paleozoic contact at site Sg5. Most of these samples are characterized by well-defined, univectorial decay to the origin in demagnetization diagrams (Figures 9e and 9f). This magnetization is found in both low- and high-coercivity magnetic phases. In thermal demagnetization, distributed laboratory unblocking temperatures up to 670°C were observed. This southeast direction is found in 63% of the samples from site Sg5 but is found only as isolated samples from the other three sites. In situ site mean directions yield a mean direction of $D = 154.9°$, $I = -16.0°$ ($k = 90.6$, $\alpha_{95} = 9.7°$) (Figure 8d and Table 3).

Paleomagnetic data and rock magnetic experiments show that Sherman Granite samples with the characteristic magnetization have distinctly different paleomagnetic and rock magnetic characteristics from those with the southeast declination remanence. Sample La44-Ja (Figure 10a), which yields the characteristic magnetization, shows rapid acquisition of IRM in fields less than 150 mT, indicating the dominance of a low-coercivity phase, probably magnetite. The slight progressive increase in IRM at higher fields indicates a small amount of hematite is present and contributes to the IRM. AF demagnetization characteristics and maximum unblocking temperatures below 580-585°C for stable remanence indicate that magnetite is the principal magnetic carrier in most granitic samples, including the red granite facies. Hematite, where present, does not appear to contribute to a stable, recoverable remanence. By contrast, a sample from site Sg5 with the southeast, negative inclination direction, shows rapid acquisition of IRM up to 150 mT, followed by gradual, essentially linear acquisition up to a peak field of 1200 mT (Figure 10b). Such behavior indicates that, although some magnetite is present, hematite is the dominant magnetic phase. AF and thermal demagnetization behavior by other samples that contain this secondary magnetization indicate that both magnetite and hematite contribute to this remanence.

$^{40}Ar/^{39}Ar$ Results and Interpretation

To evaluate the cooling and subsolidus thermal history of the Middle Proterozoic rocks from the Laramie Range, one hornblende, two biotite, and two microcline separates from the Sherman Granite were analyzed by the $^{40}Ar/^{39}Ar$ incremental step-heating method. The dated samples were collected from the same outcrops as sites La42 and La44 located in the southern Laramie Range (Figure 1).

$^{40}Ar/^{39}Ar$ results from the hornblende separate yield an total gas age of 1414 ± 3 Ma (1σ) (Table 4). Low-temperature steps are characterized by apparent argon loss with ages that increase progressively from 550 to 1400 Ma. Potassium/calcium (K/Ca) ratios from the low-temperature steps are variable and may reflect degassing of

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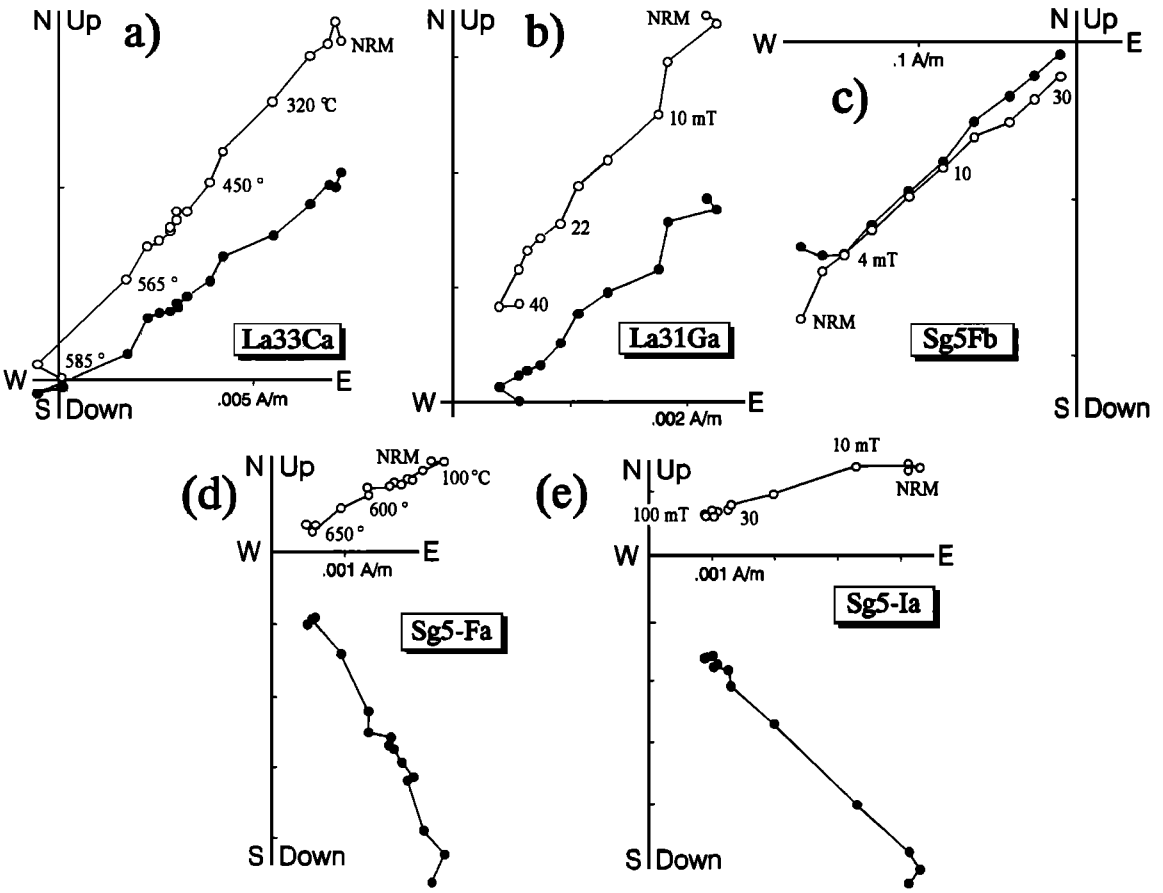


Figure 9. Orthogonal vector diagrams showing alternating field and thermal demagnetization behavior of samples from Sherman Granite and related igneous rocks. Samples La33Ca and La31Ga are from granite and show the characteristic magnetization; sample Sg4Fb is from a diorite obtained from a zone of comingled diorite and quartz monzonite in the Virginia Dale ring dike complex and has a direction anitipodal to the characteristic direction. Samples Sg5Fa and SG5Ia are from a site in the Sherman Granite near the Precambrian/Paleozoic nonconformity. These samples are suspected of having been remagnetized in late Paleozoic time. Conventions are as in Figure 5.

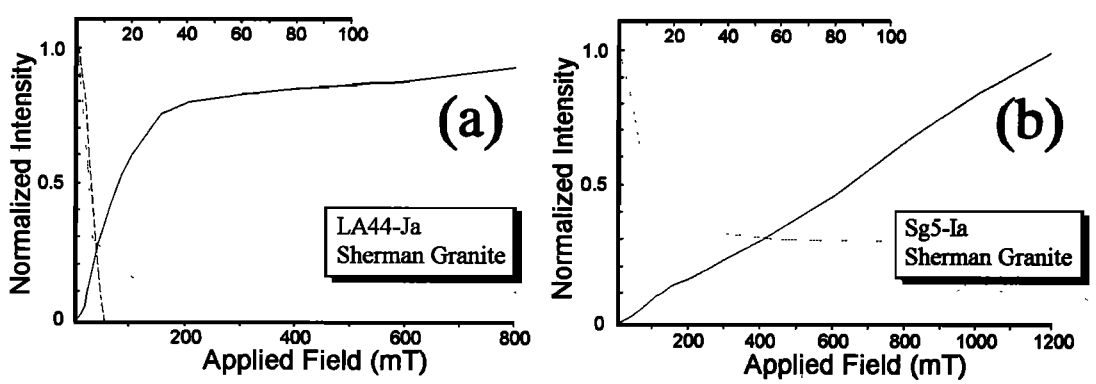


Figure 10. Normalized intensity diagrams of Sherman Granite samples showing the acquisition of isothermal remanent magnetizations (IRM) (solid line; lower scale) induced by the application of an applied direct current field, alternating field (AF) demagnetization of natural remanent magnetization (dotted line; upper scale); and AF demagnetization of IRM (dashed line; upper scale). (a) Sherman Granite sample that contains the characteristic magnetization common to the Laramie anorthosite complex and Sherman Granite; (b) Sherman Granite sample from site Sg5 located just below the Precambrian/Paleozoic nonconformity; this sample is inferred to have been remagnetized in late Paleozoic time.

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loosely held argon from the hornblende structure or slight contamination by other relatively potassium-rich phases. Intermediate- and high-temperature steps produce an essentially flat age spectrum (Figure 11a), with corresponding internally consistent K/Ca ratios. A plateau date calculated for the 900° to 1000°C steps is 1421 ± 6 Ma (2σ) and corresponds to 86% of the $^{39}Ar_K$ released. A $^{39}Ar/^{40}Ar$ versus $^{36}Ar/^{40}Ar$ inverse correlation age, based on the 750° to 1025°C steps, is 1422 +3, -7 Ma and initial $^{40}Ar/^{36}Ar$ ratio of 293 +85, -53 (mean square of weighted deviates = 1.4); a conventional $^{40}Ar/^{36}Ar$ versus $^{39}Ar/^{36}Ar$ isochron yields essentially identical results. The $^{40}Ar/^{36}Ar$ initial ratio is imprecisely defined due to the high radiogenic yields of steps used in calculation of the inverse isochron. Consequently, we cannot demonstrate the presence or absence of excess ^{40}Ar. In the absence of demonstrable excess ^{40}Ar and the apparent agreement of the observed $^{40}Ar/^{39}Ar$ initial value with the atmospheric ratio of 295.5, we consider the plateau date a reasonable estimate for the cooling age of the hornblende. This age is somewhat younger than the zircon U-Pb emplacement age reported for the Red Mountain pluton of the Laramie anorthosite complex [*Frost et al.*, 1988; *Frost et al.*, 1990], but it overlaps the

Table 4. $^{40}Ar/^{39}Ar$ Age-Spectrum Data From Sherman Granite Samples, Southern Laramie Range, Wyoming

Temperature °C		$^{40}Ar_R$	$^{39}Ar_K$	F	$^{39}Ar/^{37}Ar$	$\%^{40}Ar_R$	$\%^{39}Ar$	Apparent Age (Ma at 1σ)
Sample: La42; Hornblende; 51.1 mg; Measured $^{40}Ar/^{36}Ar_a$ = 298.9; J Value = 0.024914 ± 0.25% (1σ)								
500		0.71626	0.04983	14.37	0.93	63.0	1.1	552 ± 6
600		0.84673	0.02143	39.51	0.26	64.6	0.5	1236 ± 6
700		1.8860	0.04044	46.64	0.66	91.9	0.9	1391 ± 3
750		1.39270	0.02941	47.35	0.77	93.9	0.7	1406 ± 4
800		1.5118	0.03152	47.96	0.53	96.1	0.7	1418 ± 4
850		6.06365	0.12309	49.26	0.38	98.6	2.7	1445 ± 2
900	P	85.087	1.7670	48.15	0.34	99.7	39.2	1422 ± 3
950	P	81.349	1.6924	48.07	0.33	99.8	37.5	1420 ± 3
975	P	8.5205	0.17699	48.14	0.33	99.7	3.9	1422 ± 3
1000	P	11.141	0.23221	47.98	0.32	99.7	5.1	1419 ± 3
1025		14.997	0.30930	48.49	0.31	99.6	6.9	1429 ± 3
1050		0.93778	0.01923	48.7	0.25	98.3	0.4	1434 ± 7
1100		0.66735	0.01390	48.01	0.27	94.2	0.3	1419 ± 10
1150		0.20380	0.00393	51.9	0.36	82.0	0.1	1497 ± 53
Total gas				47.7				1414 ± 3
Sample: La42; Biotite; 15.2 mg; Measured $^{40}Ar/^{36}Ar_a$ = 299.5; J Value = 0.024805 ± 0.25% (1σ)								
500		10.447	0.35008	29.84	58.6	98.7	4.6	999 ± 2
600		28.376	0.60861	46.62	151.7	99.5	8.1	1386 ± 3
700		94.882	1.9742	48.06	335.0	99.5	26.1	1416 ± 3
750		59.582	1.2343	48.27	186.8	99.7	16.3	1420 ± 3
775		20.668	0.42663	48.44	89.7	99.5	5.6	1424 ± 3
800		19.032	0.39046	48.74	52.5	99.8	5.2	1430 ± 3
825		19.760	0.40378	48.94	29.3	99.7	5.3	1434 ± 3
850		20.969	0.42890	48.89	15.1	99.8	5.7	1433 ± 3
900		44.008	0.90224	48.78	6.4	99.9	11.9	1431 ± 3
950		27.625	0.57215	48.28	21.8	99.9	7.6	1421 ± 3
1000		5.7308	0.12041	47.60	5.7	99.2	1.6	1406 ± 3
1050		3.5513	0.07398	48.01	1.1	98.9	1.0	1415 ± 3
1200		3.2352	0.06722	48.13	2.4	98.1	0.9	1417 ± 3
Total gas				47.38				1402 ± 3
Sample: La44; Biotite; 12.4 mg; Measured $^{40}Ar/^{36}Ar_a$ = 298.9; J Value = 0.02425 ± 0.25% (1σ)								
500		1.6831	0.11625	14.48	2.3	89.9	2.3	543 ± 1
600		6.9585	0.19148	36.34	82.43	98.0	3.8	1140 ± 2
700		29.388	0.60224	48.80	69.42	98.4	11.8	1409 ± 3
750		26.414	0.53943	48.97	48.58	99.5	10.6	1412 ± 3
800		19.765	0.40090	49.30	59.01	99.8	7.9	1419 ± 3
850		19.484	0.39272	49.61	57.02	99.8	7.7	1425 ± 3
900		32.00	0.63865	50.12	58.93	99.4	12.6	1435 ± 3
950		45.833	0.91558	50.06	45.86	99.7	18.0	1434 ± 3
1000		37.81	0.76762	49.26	23.66	99.8	15.1	1418 ± 3
1050		15.03	0.30685	48.98	19.31	99.4	6.0	1412 ± 3
1150		9.6291	0.19566	49.21	9.47	98.4	3.8	1417 ± 3
1300		0.86214	0.01772	48.64	3.20	71.2	0.3	1406 ± 9
Total gas				48.16				1396 ± 3

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Table 4. (Continued)

Temperature °C	$^{40}Ar_R$	$^{39}Ar_K$	F	$^{39}Ar/^{37}Ar$	$\%^{40}Ar_R$	$\%^{39}Ar$	Apparent Age (Ma at 1σ)
Sample: La42; Microcline; 12.1 mg; Measured $^{40}Ar/^{36}Ar_a$ = 299.5; J Value = 0.024395 ± 0.25% (1σ)							
500	1.5897	0.03640	43.67	36.9	95.1	0.4	1308 ± 5
600	0.86075	0.03706	23.22	----	98.1	0.4	810 ± 5
700	6.5988	0.24598	26.82	50.4	94.5	2.8	908 ± 2
750	11.46	0.35946	31.88	80.7	99.4	4.1	1038 ± 2
800	27.865	0.81636	34.13	150.8	99.6	9.2	1093 ± 2
850	17.362	0.48620	35.71	107.0	99.7	5.5	1130 ± 2
900	15.030	0.41600	36.13	87.0	99.5	4.7	1140 ± 3
950	14.216	0.40136	35.42	----	99.6	4.5	1124 ± 2
1000	19.251	0.53211	36.18	95.5	99.7	6.0	1141 ± 2
1050	23.883	0.64718	36.90	69.4	99.7	7.3	1158 ± 2
1100	21.739	0.57376	37.89	75.5	99.9	6.5	1181 ± 2
1200	91.610	2.3707	38.64	95.6	99.9	26.9	1198 ± 2
1400	75.151	1.9048	39.45	75.7	99.7	21.6	1216 ± 3
Total gas			37.00				1160 ± 2
Sample: La44; Microcline; 12.7 mg; Measured $^{40}Ar/^{36}Ar_a$ = 298.9; J Value = 0.023945 ± 0.25% (1σ)							
500	2.2974	0.05663	40.57	45.36	93.9	0.6	1224 ± 3
600	4.0887	0.15097	27.08	168.2	98.3	1.6	902 ± 2
700	18.236	0.56145	32.48	125.6	97.7	5.9	1038 ± 2
750	21.455	0.59969	35.78	79.7	99.6	6.3	1116 ± 2
800	23.079	0.62501	36.93	74.3	99.6	6.6	1143 ± 2
850	19.840	0.52698	37.65	100.1	99.5	5.5	1159 ± 2
900	18.248	0.47813	38.17	68.9	99.4	5.0	1171 ± 2
950	25.498	0.65539	38.91	109.5	99.5	6.9	1188 ± 2
1000	45.773	1.1483	39.86	93.6	99.6	12.0	1209 ± 2
1100	28.995	0.71637	40.48	85.6	99.5	7.5	1222 ± 3
1150	56.128	1.3713	40.93	129.0	99.6	14.4	1232 ± 3
1250	94.069	2.3292	40.39	81.7	99.7	24.4	1220 ± 3
1400	13.081	0.31742	41.21	70.6	98.2	3.3	1239 ± 3
Total gas			38.88				1187 ± 2

The ratio $^{40}Ar/^{36}Ar_a$ is measured atmospheric argon used for mass discrimination value at time of analysis; $^{40}Ar_R$ is radiogenic ^{40}Ar; $^{39}Ar_K$ is potassium-derived ^{39}Ar; F is the ratio of $^{40}Ar_R$ to $^{39}Ar_K$ after correction for mass discrimination and interfering isotopes; $^{39}Ar/^{37}Ar$ is the ratio of $^{39}Ar_K$ to $^{37}Ar_{Ca}$ (this value can be converted to the approximate K/Ca by multiplying by 0.5); $\%^{40}Ar_R$ and $\%^{39}Ar$ are percent of radiogenic ^{40}Ar and percent of total ^{39}Ar released in each temperature step; P denotes steps used in the calculation of the plateau age for hornblende La42. Abundances of $^{40}Ar_R$ and $^{39}Ar_K$ are reported in volts. Conversion of moles can be made by using a conversion factor of 9.736×10^{-13} moles argon per volt of signal.

range of U-Pb zircon dates from the Sherman Granite reported by *Aleinikoff* [1983]. This age probably reflects relatively rapid cooling of the granite after emplacement to temperatures below 500 ± 25°C, the closure temperature for diffusion of intracrystalline argon [*Harrison*, 1981].

The biotites from La42 and La44 yield statistically identical (at the 95% confidence level) total gas dates of 1402 ± 3 Ma and 1396 ± 3 Ma (Table 4). Both age spectra (Figure 11b) are similar and show apparent argon loss in low-temperature steps, followed by an abrupt climb to ages greater than 1400 Ma at intermediate- and high-temperature steps. Both spectra are internally discordant, displaying a convex up or "hump-shaped" release (Figure 11b). Maximum ages are reached at intermediate temperatures; this is followed by an overall, slightly progressive decline in apparent age (Table 4). The maximum apparent ages recorded by both biotites are greater than that of the hornblende plateau age described above. Similar disturbed age spectra for biotite have been described by several workers [*Tetley and McDougall*, 1978; *Foland*, 1983; *York and Lopez-Martinez*, 1986; *Hubacher and Lux*, 1987; *Heizler et al.*, 1988] and may be caused by the presence of small amounts of alteration phases such as chlorite, which do not retain ^{40}Ar in a quantitative fashion or by loss of ^{39}Ar from recoil during irradiation [*Hess and Lippolt*, 1986]. Petrographic examination of the freshest Sherman Granite samples indicates that biotite commonly shows evidence of incipient to slight chloritic alteration. Despite efforts to obtain pure separates by hand-picking, some chlorite may be present in biotite at the subgrain scale. Because the closure temperature to ^{40}Ar diffusion of hornblende is approximately 500°C, whereas that for biotite ranges from 280°to 345°C depending upon cooling rate [*McDougall and Harrison*, 1988], the maximum apparent ages are clearly anomalous. Because of the very high radiogenic yields of most biotite steps (>99%), isotope correlation diagrams or isochrons did not yield useful estimates of $^{40}Ar/^{36}Ar$ initial ratios; hence the presence of excess ^{40}Ar cannot be evaluated. Consequently, we take the total gas dates of the biotites to represent a minimum estimate of the time of biotite closure to ^{40}Ar diffusion.

Sherman Granite microclines yield similar age spectra (Figure 11c) and total gas ages of 1160 ± 2 Ma (La42) and 1187 ± 2 Ma (La44). Except for initial temperature steps that record old apparent

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ages (i.e., 0.5 to 1.0% of the total $^{39}Ar_K$ released), the age of individual gas fractions increases in a convex-up manner to broad plateau-like areas that represent about 50% of the gas released, with average apparent ages of 1205 and 1239 Ma. Individual temperature steps are relatively constant with respect to volume of $^{39}Ar_K$ released at each temperature step up to temperatures of 1000° to 1100°C, where proportionately larger fractions are released; this enhanced outgassing may reflect a profound change in diffusion mechanism or may be caused by incongruent melting of the potassium feldspar. The shape of the age spectra (Figure 11c) is similar to theoretical curves produced for microclines of planar geometry that have experienced slow cooling (approximately 5°C/m.y.) through a range of closure temperatures ranging from 130° to 160°C [*Harrison and McDougall*, 1982]. Theoretical curves for slow cooling and episodic argon loss are similar, however, and it is not possible to uniquely distinguish between the two models on the basis of individual age spectra [*Harrison and McDougall*, 1982].

The $^{40}Ar/^{39}Ar$ geochronologic data from our study and published U-Pb and K-Ar dates (summarized in Table 1) have been combined to illustrate the emplacement and cooling history of the Sherman Granite and Laramie anorthosite complex (Figure 12). The "closure temperature" for zircon is not known with certainty, although assigned temperatures range from 700°C [*Cumming and Krstic*, 1991; *Mattinson*, 1978; *Harrison et al.*, 1987; *Ghent et al.*, 1988] to greater than 1000°C [*Cliff*, 1985], with temperatures above 750°C commonly cited. For this study, we have assigned a minimum closure temperature of 760°C for the Sherman Granite, and a temperature of 800°C has been assigned to the Red Mountain pluton. Closure to ^{40}Ar diffusion in hornblende ranges from 580° to 480°C, depending on cooling rate [*Harrison*, 1981; *McDougall and Harrison*, 1988]; for the Sherman Granite hornblende, we have used a temperature of 500°C. Closure temperatures for biotite range from 350° to 280°C [*Harrison et al.*, 1985] and we assigned a minimum closure temperature of 300°C. Determination of appropriate closure temperatures for the microclines is more problematic. Recent studies have demonstrated that potassium feldspars may contain multiple ^{40}Ar diffusion domains that record a range of closure temperatures [*Lovera et al.*, 1989]. However, because we have not performed appropriate degassing experiments necessary to identify and quantify diffusion domains and closure temperatures using Arrhenius relationships, we have assigned maximum and minimum closure temperatures of 165° and 130°C, respectively.

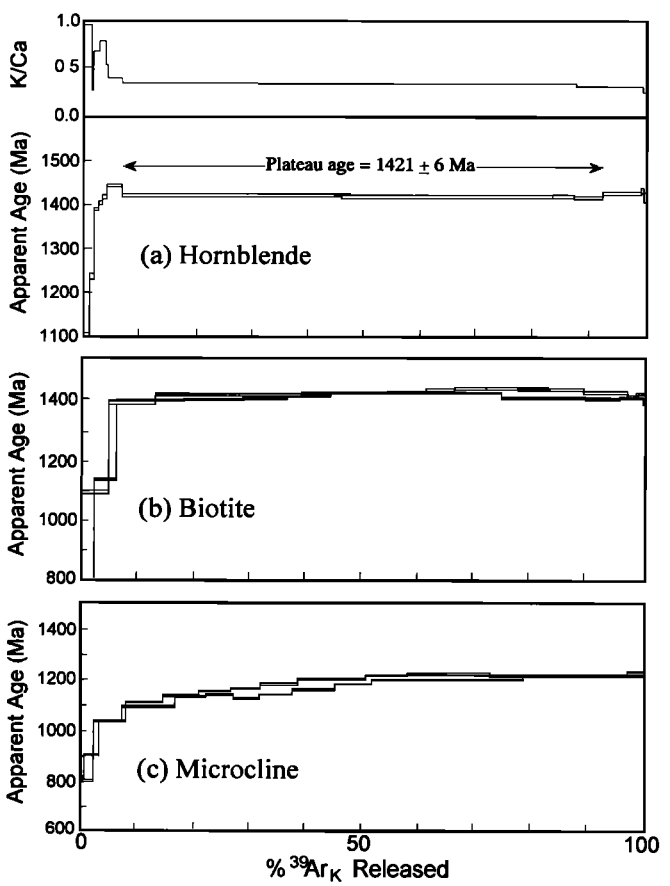


Figure 11. $^{40}Ar/^{39}Ar$ apparent age spectra of hornblende, biotite, and microcline from granitic samples of the Sherman Granite, southern Laramie Range, Wyoming. Also shown is the apparent K/Ca spectra for the hornblende sample. Width of the horizontal bar in each diagram is the analytical error at ± 1σ; the analytical error in the hornblende plateau age is given at ± 2σ. Open error bars are samples from site La42, solid error bars are from site La44.

The $^{40}Ar/^{39}Ar$ geochronologic data presented here, combined with previous isotopic dating (summarized in Table 1 and shown graphically in Figure 12), indicate that the Sherman Granite (and by inference the Laramie anorthosite complex) experienced a relatively simple tectonothermal history since emplacement at about 1435 Ma. The U-Pb emplacement age of 1431 ± 6 Ma for the Sherman Granite and the hornblende cooling age of 1421 ± 6 Ma (2σ) are statistically indistinguishable at the 95% confidence level, whereas the biotites give total gas dates of about 1400 Ma. Using closure temperatures discussed above, these data are interpreted to record moderate to rapid cooling (25 to 260°C/m.y.) through hornblende closure after emplacement, followed by an exponential decline in cooling rate as recorded by biotite and microcline apparent ages. Taken at face value, the difference between biotite and hornblende ages indicate slow cooling (6 to 26°C/m.y.) between about 500° and 300°C, although we caution that because the biotite ages probably represent minimum ages, cooling rates may have been more rapid.

Such relatively slow cooling rates are surprising given that quantitative thermal models for conductive cooling of plutons predict cooling to near ambient temperatures, even for large plutons, should occur in less than 1 to 4 m.y. [*Cathles*, 1981; *Spera*, 1980]; if convection occurs, cooling takes significantly less time. Such models are inherently simplistic, however, and are usually based on single intrusions of simple geometry. They may not be applicable to complex, potentially long-lived, multiple intrusions such as the Sherman Granite. Recently, for example, *Chesley et al.* [1993] demonstrated that magmatism in the Carboniferous Cornubian batholith of southwest England spanned 20 to 25 m.y., and emplacement of individual plutons occurred over 5 to 10 m.y. Similarly, U-Pb dating indicates that magmatism in the Proterozoic Pikes Peak batholith of central Colorado extended over 30 m.y. (*D. Unruh*, personal communication, 1992). Such long-lived magmatism may perturb and elevate local geotherms, thus retarding cooling of the batholith, and may contribute to low apparent cooling rates such as observed in this study. Alternatively, widespread crustal extension and anorogenic magmatism at 1400 Ma may have also contributed to slow apparent cooling rates by elevating regional geotherms. Further work is necessary in order to more fully define the thermal history of the Sherman Granite and Laramie anorthosite complex. Tentatively, we conclude that cooling through hornblende and biotite closure temperatures occurred within 10 to 20 m.y. of emplacement, and we note that neither the hornblende nor biotite age spectra show evidence of significant thermal disturbance since closure at about 1420 to 1400 Ma.

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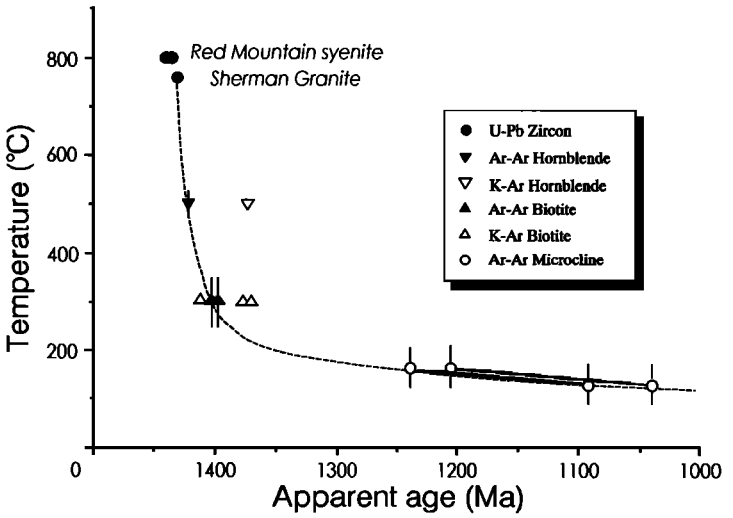


Figure 12. Cooling curve illustrating the thermal history of the southern Laramie Range based on isotopic dating of the Sherman Granite and Laramie anorthosite complex. Construction of the cooling curve is based solely on U-Pb and $^{40}Ar/^{39}Ar$ dates; other isotopic dates are shown for reference only. Assigned closure temperatures for each mineral are described in the text. The vertical line in the $^{40}Ar/^{39}Ar$ dates represents the uncertainty in closure temperature.

The microcline $^{40}Ar/^{39}Ar$ age spectra provide important information regarding the low temperature history of both intrusions. The microclines give similar maximum ages of 1202 Ma (La42) and 1238 Ma (La44). We interpret the microcline age spectra as recording continuous or nearly continuous closure to ^{40}Ar diffusion through temperatures of 165° to 130°C [*Harrison and Be*, 1983] at 1200 to 900 Ma (Figure 12). These temperatures may reflect slow cooling and/or uplift of the Sherman Granite in late Middle Proterozoic time. Alternatively, these data may record a low-temperature thermal event at about 1000 Ma that partially reset the microcline system but did not significantly affect either the biotites or hornblende. Distinguishing between the two explanations is not possible based on the feldspar data, as both could result in similar age spectra. At present, however, there is no geologic or geochronologic evidence for circa 1000 Ma deformation and/or thermal disturbance in southern Wyoming or northern Colorado.

Discussion of Paleomagnetic Results

Age and Origin of the Characteristic Magnetization

The well-defined characteristic magnetizations of the Laramie anorthosite complex and Sherman Granite were probably acquired during subsolidus cooling following emplacement at about 1435 Ma. Three lines of evidence support this hypothesis.

First, the characteristic magnetizations of both plutons (Figures 4c and 8c and Table 3) are similar and the combined mean direction is 120° removed from the present-day field and 115° removed from the Recent time-averaged direction. Furthermore, the characteristic direction is distinctly different from known directions for the Late Proterozoic or any part of the Phanerozoic. Thus, barring significant structural disturbance, postemplacement whole-scale remagnetization of both plutons is unlikely.

Second, the $^{40}Ar/^{39}Ar$ dates and previously published isotopic dates [*Zielinski et al.*, 1981; *Aleinikoff*, 1983] indicate that the Sherman Granite and the Laramie anorthosite complex experienced a relatively simple thermal history since intrusion at about 1435 Ma. Cooling through ^{40}Ar closure temperatures for biotite and hornblende was relatively rapid and occurred within a maximum of 10 to 20 m.y. following emplacement (Figure 12). A thermal event that may have caused either partial of complete remagnetization appears unlikely. With the exception of a few sites in the Sherman Granite located near the Precambrian/Paleozoic unconformity that carry a magnetization of inferred late Paleozoic age, there is no record of stable secondary magnetizations of younger age.

Third, the remanence of the Laramie anorthosite and associated syenites is extremely stable during demagnetization and is carried by magnetic phases characterized by a high, narrow range of laboratory unblocking temperatures and high median destructive fields. Rock magnetic data and STEM observations suggest that this remanence is carried by elongate to rod-shaped, low-Ti titanomagnetite inclusions in plagioclase or potassium feldspar. The rock magnetic properties displayed by most samples are characteristic of single-domain magnetite. The occurrence of single-domain magnetite in feldspar as the carrier of stable remanence in a variety of rock types has been described by numerous workers [*Scofield and Roggenthen*, 1986; *Morgan and Smith*, 1981]. The temperature at which magnetite exsolution occurs is not known. *Morgan and Smith* [1981] suggested that knowledge of the temperature at which exsolution occurs is needed to determine whether the observed remanence is of thermoremanent or chemical remanent magnetization origin and to properly interpret the observed range of laboratory unblocking temperatures. *Armbrustmacher and Banks* [1974] have suggested that exsolution of magnetite in some rocks may occur at temperatures less than 475°C. In the Laramie anorthosite complex, remanence acquisition clearly postdated doming, but the temperature at which this occurred is unknown. The characteristic magnetization of the Laramie anorthosite complex may be a combination of thermoremanent and high-temperature (>400°C) thermochemical magnetizations acquired during subsolidus cooling of the complex.

By contrast, stable magnetization in Sherman Granite is of low to moderate coercivity and characterized by a broad range of unblocking temperatures below 585°C. Consequently, remanence in the granite is carried primarily by magnetite of a distributed range of grain sizes and/or domain states. This conflicts with the findings of *Eggler and Larson* [1968], who reported that finely dis-

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seminated hematite carried the magnetization in the red facies of Sherman Granite. We found no rock magnetic evidence to corroborate this conclusion.

The similarity of remanence in both plutons, carried by magnetic phases of significantly different rock magnetic characteristics, and lack of similarity of the characteristic directions with magnetizations of younger age, provide strong evidence that the observed remanent magnetizations in both plutons were probably acquired during cooling. The age of Sherman Granite magnetization can be closely approximated as 1415 Ma. The age of the Laramie anorthosite complex magnetization may be somewhat older, based on its slightly greater U-Pb crystallization age, but the mean directions of both plutons are indistinguishable at the 95% confidence level; thus the combined pole is also assigned an age of 1415 Ma.

Paleomagnetic Poles and Analysis of Secular Variation

Paleomagnetic poles for the Laramie anorthosite complex and Sherman Granite have been calculated, assuming no posttectonic tilt or vertical axis rotation, by averaging site mean VGPs (Table 3). In Figure 13, we plotted only poles calculated after excluding sites with α_{95} values greater than 20° or sites having less than two samples of the characteristic direction. Although this selection criteria may appear lenient, adoption of more stringent criteria does not appreciably change this result. Also shown in Figure 13 are previously determined poles for the Laramie anorthosite complex (using the 12.0°S, 201.0°E position) and Sherman Granite [*Larson et al.*, 1973; *Eggler and Larson*, 1968]. Both poles from this study plot northeast of and indicate lower paleolatitudes than the previous ones. The Sherman Granite pole from this study is displaced slightly northwest of the new Laramie anorthosite complex pole, although both are statistically indistinguishable at the 95% confidence level.

Combining the paleomagnetic data from the Sherman Granite and the Laramie anorthosite complex yields 37 individual VGPs with a corresponding paleomagnetic pole at 215.0°E longitude and 6.7°S latitude (Figure 13) with $A_{95} = 3.5°$ and an angular standard deviation (δ_{63}) of 11.9° (Table 3). Upper and lower confidence limits of 2.3° and 1.7° for δ_{63} were calculated using the method of *Cox* [1969]. The expected VGP scatter (S_λ) for 29.5° latitude (the calculated paleolatitude of the southern Laramie Range) using Model G of *McFadden et al.* [1991] for the interval 5-0 Ma is 14.5 ± 1.0°. Our observed value is slightly less than predicted, but the confidence intervals overlap. Assuming recent models of paleosecular variation are appropriate for the Precambrian, the relatively low δ_{63} may indicate that the VGPs from the Laramie anorthosite complex and the Sherman Granite do not average secular variation. We consider this unlikely for the following reasons. First, both the Laramie anorthosite complex and the Sherman Granite are relatively large and composite plutons, and cooling must have taken longer than the minimum 10^5 years assumed necessary to average secular variation. Geochronologic data from this study indicate that cooling through appropriate magnetic blocking temperatures (approximately 580° to 350°C) may have occurred as slowly as 10°-30°/m.y. suggesting that sufficient geomagnetic variation may have been recorded by both plutons. Second, antiparallel magnetizations in three sites in the Laramie anorthosite complex indicate that sufficient averaging of secular variation may have occurred at the site level. Paleomagnetic studies from intrusions from the Sierra Nevada batholith [*Frei et al.*, 1984] and the southern Cascades [*Dodson et al.*, 1978] demonstrate that plutonic rocks, even those with relative high and narrow ranges of laboratory unblocking temperatures, can average secular variation even at the site level, and that slow cooling can result in estimates of VGP scatter significantly less than predicted. Finally, antipodal magnetizations are present in Sherman granite and redated rocks indicating that remanence acquisition spanned at least two polarity chrons. We conclude that the combined VGPs from the Laramie anorthosite complex and Sherman Granite study probably average secular variation and provide a reliable recording of the geomagnetic field at about 1415 Ma.

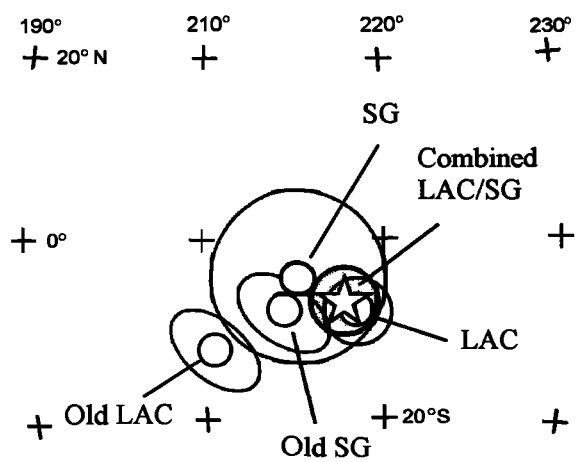


Figure 13. Equal area projection centered on 0° latitude, 210°E longitude, showing the location of paleomagnetic poles determined from the Laramie anorthosite complex (LAC), Sherman Granite (SG), and combined LAC/SG pole (star). Also shown are previously published virtual geomagnetic poles for the LAC [*Eggler and Larson*, 1968] and SG [*Larson et al.*, 1973].

Uncertainties Regarding Application of Tilt Corrections

The tilt of the Laramie anorthosite complex and Sherman Granite represents the greatest uncertainty in this study, as it does with all paleomagnetic investigations of plutonic rocks. Three recognizable events that may have produced tilt are (1) uplift and exposure of Precambrian rocks prior to sedimentation in the early Paleozoic, (2) possible tilting caused by late Paleozoic uplift of the ancestral Rocky Mountains, and (3) Laramide foreland deformation in the Late Cretaceous or early Tertiary. Laramide uplift is arguably the most likely cause of tilt in the southern Laramie Range, but none of these can be corrected for because Paleozoic and younger strata have been stripped from the core of the range and are absent in most areas we sampled. At two sites where Pennsylvanian sedimentary rocks are present or nearby (La30 and Sg5, Figures 1 and 2), the strata dip less than 5°, and a tilt correction has not been applied to the site mean data or VGPs.

To investigate the effects of unrecognized tilt on the Laramie anorthosite complex/Sherman Granite pole, we corrected our data to account for tilts of 10° and 20° about a north trending axis (the approximate trend of thrust faults that bound the Laramie Range). Because the Laramie Range was interpreted to be underlain by west dipping thrust faults (Brewer et al., 1982), west-side-down tilt is probably the most appropriate structural correction for these rocks and is consistent with observations of *Brewer et al.* [1982] that Pennsylvanian sedimentary strata dip gently off the western edge of the mountains. Corrections for either west or east tilt result in shift of the pole along a small circle subparallel to the equatorial axis (Figure 14a), away from the locus of paleomagnetic poles of similar age from elsewhere in North America. Given the uncertainties

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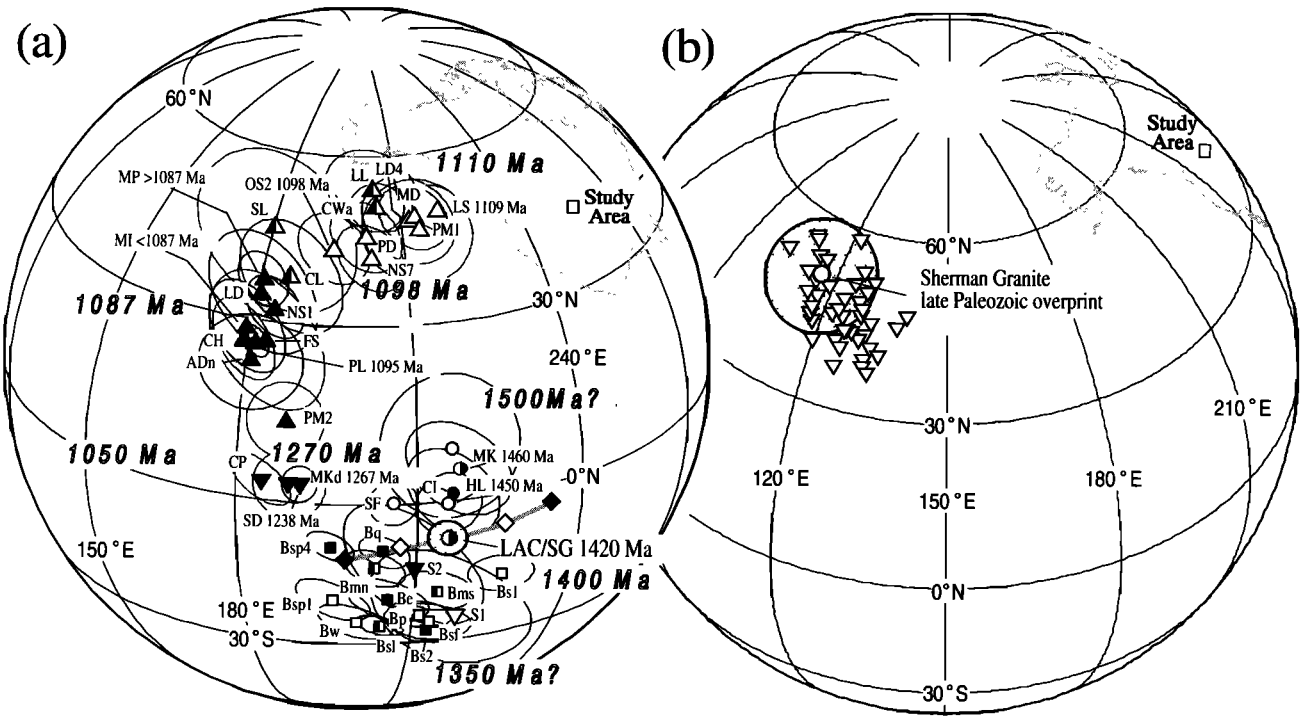


Figure 14. (a) Orthographic global projection centered on 25°N, 200°E, showing the location of the combined Laramie anorthosite complex/Sherman Granite pole and paleomagnetic poles and 95% confidence limits from 1500 to 1090 Ma igneous and sedimentary rocks from elsewhere in North America. Circles show poles derived from igneous intrusions with ages of 1480 to 1415 Ma; squares show poles from sedimentary strata of the Belt Supergroup; downward pointing triangles are from the Sibley Group and volcanic rocks and dikes in the Northwest Territories; and upward pointing triangles are poles from the Midcontinent Rift and central Arizona that define the Keweenawan track. Solid symbols denote poles derived from normal polarity paleomagnetic data; open circles are poles from reverse polarity data; and half-solid symbols represent mixed polarity. Abbreviations for poles, pole locations, geochronologic data, and appropriate references are listed in Table 5. The heavy gray line shows the inferred track of apparent polar wander from about 1500 to 1090 Ma. The thin gray line shows the possible effects of unrecognized tilts of 10° (open diamonds) and 20° (solid diamonds) about a north trending axis on the position of the Laramie anorthosite complex/Sherman Granite pole. West tilt moves the pole southeast; east tilt causes northeast movement. In either case, the pole is moved away from the main cluster of 1480 to 1450 Ma poles. (b) Orthographic global projection centered on 40°N, 150°E, showing paleomagnetic poles from rocks in North America that are known or suspected to have been remagnetized during late Paleozoic time. Also shown is the mean VGP and A_{95} from four sites in Sherman Granite that were remagnetized during the late Paleozoic. Late Paleozoic poles are from the compilation of *McCabe and Elmore* [1989].

described above, and the possibility that Laramide structures vary significantly along the trend of the Laramie Range [*Brewer et al.*, 1982], we cannot realistically apply tilt corrections to our data and must rely on in situ paleomagnetic directions for pole calculations. The possibility of local vertical-axis rotation of the southern Laramie Range cannot be assessed and must be assumed to be negligible.

Middle Proterozoic North American APW Path

In order to compare paleomagnetic results from the Laramie anorthosite complex and Sherman Granite with rocks of similar age elsewhere in North America we have constructed a Middle Proterozoic APW path for the interval 1500 to 1090 Ma (Figure 14a) using poles listed in Table 5 (most having a reliability index $Q \geq 3$ [*Van der Voo*, 1989]). Wherever possible, the age of segments of the APW path in Figure 14a have been calibrated using high-precision U-Pb or $^{40}Ar/^{39}Ar$ dates.

Most North American paleomagnetic poles for the interval 1500 to 1400 Ma are derived from intrusions or volcanic rocks associated with widespread plutonism that occurred throughout the Laurentian craton [*Anderson*, 1983]. In general, these poles are relatively well grouped and plot slightly north of the equator in the central Pacific ocean (Figure 14a). Isotopic dates for intrusions from which the poles are derived give apparent ages that range from 1480 to 1450 Ma (Table 5). The slightly younger pole of the Laramie anorthosite complex/Sherman Granite is located somewhat south of the main clustering of poles, but is statistically indistinguishable from poles of the Croker Island and Harp Lake complexes of Labrador. Assuming none of the poles have been tilted significantly, the overall agreement of poles from Labrador to Wyoming indicates that there has been little or no appreciable differential rotation between these areas since Middle Proterozoic time as originally noted by *Murthy et al.* [1968] and that APW from 1480 to 1415 Ma was minimal. The overall agreement between the Laramie anorthosite complex/Sherman Granite pole and those from anorogenic intrusions elsewhere in North America is consistent with an age of magnetization of about 1415 Ma and with the conclusion that the southern Laramie Range probably has not experienced significant tilting due to Laramide deformation.

All poles from circa 1480 to 1415 Ma igneous rocks, including that of the Laramie anorthosite complex/Sherman Granite, plot sig-

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Table 5. Age, Paleomagnetic Data, and Quality Index for Paleomagnetic Poles Used in the Middle Proterozoic Apparent Polar Wander Path (Figure 14a)

Rock Unit, Location	Symbol*	Age, Ma	Method	Plat.	Plong.	δp	δm	A_{95}	Reliability 1 2 3 4 5 6 7	$Q^{\dagger}$	Reference
Michikamau anorthosite, Labrador	MK	1460	U-Pb, zr	1.5°S	218.0°E	--	--	4.5°	x x o o o o x	4	*Murthy et al.* [1968], *Emslie et al.* [1976]
Croker Island Complex	CI	1475	Rb-Sr	5.4°N	217.2°E	6.7°	10.5°	--	x x o o o o x	3	*Palmer* [1969]
Harp Lake anorthosite	HL	1450	U-Pb, zr	0.2°N	206.0°E	--	--	4.0°	x x o o o x x	4	*Irving et al.* [1977]
St. Francois volcanics, Missouri	SF	1480	U-Pb, zr	1.3°N	216.1°E	6.1°	9.6°		x x o o o o x	3	*Hays and Scharon* [1966], *Van Schmus et al.* [1993]
Laramie anorthosite/Sherman Granite, Wyoming	LAC/SG	1415	Ar-Ar, hb, bio	6.7°S	215.0°E	--	--	3.5°	x x x o o x x	5	*This study*
Belt SG, Spokane Fm, Montana‡	Bs1			15.5°S	225.5°E	3.7°	6.2°	--	o x x o x o x	4	*Vitorello and Van der Voo* [1977]
Belt SG, Spokane Fm, Montana	Bs2			27.2°S	212.9°E	2.8°	5.4°	--	o x o o x o x	3	*Elston and Bressler* [1980]
Belt SG, Empire Fm, Montana	Be			19.1°S	205.6°E	4.0°	7.4°	--	o x o o x o x	3	*Elston and Bressler* [1980]
Belt SG, Snowslip Fm, Montana	Bsl			25.1°S	204.4°E	2.1°	4.1°	--	o x o o x o x	3	*Elston and Bressler* [1980]
Belt SG, Shepard Fm, Montana	Bsf			24.8°S	213.2°E	4.2°	7.9°	--	o x o o x o x	3	*Elston and Bressler* [1980]
Belt SG, Wallace Fm, Montana	Bw			24.2°S	200.0°E	3.4°	6.6°	--	o o o o x o x	2	*Elston and Bressler* [1980]
Belt SG, Striped Peak Fm (Unit 1), Montana	Bsp1			19.1°S	196.2°E	3.3°	6.2°	--	o x o o x o x	3	*Elston and Bressler* [1980]
Belt SG, Mount Shields Fm, Montana	Bms			22.8°S	211.6°E	3.0°	5.6°	--	o x o o x x x	4	*Elston and Bressler* [1980]
Striped Peak Fm, Belt SG (Unit 4), Montana	Bsp4			8.5°S	195.9°E	2.3°	4.0°	--	o x o o x o x	3	*Elston and Bressler* [1980]
Belt SG, Bonner Quartzite, Montana	Bbq			9.0°S	204.7°E	2.7°	4.5°	--	o x o o x o x	3	*Elston and Bressler* [1980]
Belt SG, McNamara Fm, Montana	Bmn			12.2°S	203.2°E	4.0°	7.4°	--	o x o o x x x	4	*Elston and Bressler* [1980]
Belt SG, Pilcher Quartzite, Montana	Bp			17.1°S	214.0°E	4.0°	6.9°	--	o o o o x o x	2	*Elston and Bressler* [1980]
Lower Sibley Group, Ontario	S1			24.0°S	218.0°E	10.0°	19.0°	--	x o o o x o x	3	*Robertson* [1973a]
Upper Sibley Group, Ontario	S2			13.0°S	210.0°E	8.0°	16.0°	--	x o o o x o x	3	*Robertson* [1973a]
Coppermine River Group, Northwest Territories	CR	1267	U-Pb, bd	2.0°N	180.0°E	3.0°	6.0°	--	x x o o x o x	4	*Barager and Robertson* [1973]
MacKenzie Dikes, Northwest Territories	MKd	1267	U-Pb, bd	2.5°N	190.2°E	--	--	7.4°	x x o o x o x	4	*Irving et al.* [1972], *Park* [1974], *Schwarz and Buchan* [1982], *Le Cheminant and Heaman* [1989]
Sudbury Dikes, Ontario	SD	1238	U-Pb, bd	2.5°S	192.0°E	--	--	2.5°	x x x x o o x	5	*Palmer et al.* [1977], *Schwarz and Buchan* [1982] *Krogh et al.* [1987]
Logan sills (mean), Midcontinent Rift	LS	1109	U-Pb	49.4°N	219.9°E	3.9°	4.4°	--	x x x x o o x	5	*Halls and Pesonen* [1982], *Davis and Sutcliffe* [1985]
Marquette dikes, Midcontinent Rift	MD			48.4°N	213.5°E	5.2°	6.2°	--	x x x x o o x	5	*Pesonen and Halls* [1979]
Upper Powder Mill Group, Midcontinent Rift	PM1			46.0°N	214.0°E	--	--	9.2°	o x x x x o x	5	*Palmer and Halls* [1986]
Logan Dikes (Unit 4), Midcontinent Rift	LD4			34.7°N	180.6°E	7.4°	11.9°		x x x x o o x	5	*Pesonen* [1979]
Pukaskwa dikes, Midcontinent Rift	PD			45.2°N	201.7°E	5.1°	6.7°	--	o x x x o o x	4	*Halls and Pesonen* [1982]
North Shore Group, Midcontinent Rift	NS7			41.5°N	202.7°E	4.6°	6.4°	--	x x x x x o x	6	*Halls and Pesonen* [1982]
Osler Group (upper part), Midcontinent Rift	OS2	1098	U-Pb, bd	43.1°N	194.5°E	5.0°	6.8°	--	x x x x o o x	5	*Halls and Pesonen* [1982] *Davis and Sutcliffe* [1985]

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Table 5. (Continued)

Rock Unit, Location	Symbol*	Age, Ma	Method	Plat.	Plong.	δp	δm	A_{95}	Reliability 1 2 3 4 5 6 7	$Q^{\dagger}$	Reference
Lackner Lake Complex	LL			53.7°N	203.5°E	6.7°	8.3°	--	o x x o x o x	4	*Symons* [1989b]
Coldwell Complex§	CWa	1108	U-Pb, bd	50.4°N	204.5°E			4.4°	x x x o x x x	6	*Lewchuk and Symons* [1990], *Heaman and Machado* [1992]
Seabrook Lake Carbonatite Complex	SL			46.0°N	180.0°E	11.0°	17.0°	--	x x x x o x x	6	*Symons* [1992]
Chipman Lake Carbonatite Complex	CL			38.0°N	186.0°E	7.0°	11.0°	--	o x x x x x x	6	*Symons* [1992]
Firesand River Carbonatite Complex	FS			27.0°N	183.0°E	8.0°	13.0°	--	o x x x x o x	5	*Symons* [1989a]
Mamainse Point Volcanics, Midcontinent Rift	MP	1087		36.8°N	180.4°E	3.5°	5.3°	--	x x x x x o x	6	*Palmer* [1970], *Robertson* [1973b]
Logan dikes (mean), Midcontinent Rift	LD			48.0°N	212.0°E	7.0°	8.3°	--	x x x x o o x	5	*Halls and Pesonen* [1982]
North Shore Group, Midcontinent Rift	NS1			32.4°N	183.8°E	3.8°	6.0°	--	x x x x x o x	6	*Halls and Pesonen* [1982]
Michipicoten Island Fm, Midcontinent Rift	MI	1087	U-Pb, Zr	28.7°N	179.0°E	5.2°	8.7°	--	x x x o x o x	5	*Palmer and Davis* [1987]
Clay-Howells Carbonatite Complex	CH			26.5°N	178.8°E	4.1°	7.6°	--	x x x o o o x	3	*Lewchuk and Symons* [1989]
Portage Lake Volcanic Group (mean pole) Midcontinent Rift	PL	1095	U-Pb, bd	26.5°N	181.2°E	1.7°	2.9°	--	x x x x x o x	6	*Halls and Pesonen* [1982], *Davis and Paces* [1990]
Lower Powder Mill Group, Midcontinent Rift	PM2			14.0°N	188.0°E	--	--	6.4°	o x x x x o x	5	*Palmer and Halls* [1986]
Arizona diabase sheets, central Arizona	ADn	1090-1100	U-Pb, zr bd	22.7°N	179.3°E	--	--	7.8°	x x x o o x x	5	*Harlan* [1993]

Reliability criteria [Van der Voo, 1989] are 1, well determined rock age and a presumption that magnetization is the same age; 2, sufficient number of samples ($N > 24$), k (or K) ≥ 10 and $\leq \alpha_{95}$ (A_{95}) $\leq 16.0°$; 3, adequate demagnetization that demonstrably includes vector subtraction; 4, field tests that constrain the age of magnetization; 5, structural control, and tectonic coherence with craton or block involved; 6, the presence of reversals; 7, no resemblance to paleopoles of younger age (by more than a period); x indicates that it meets the stated critieria; o means that it fails the criteria; Q is the numerical sum of the accepted criteria. U-Pb is Uranium-Lead; Rb-Sr is Rubidium-strontium; Ar-Ar is $^{40}Ar/^{39}Ar$; zr is zircon; bd is baddelyite; hb is hornblende; bio is biotite. Other symbols are as in Tables 2 and 3.

*Symbols are given for poles in Figure 14a.

†Values for Q assigned for poles by the authors of this study. With the Exception of two poles from the Belt Supergroup, all poles used have a reliability index ≥ 3.

‡Ages assigned to Belt poles range from 1225 to 1400 Ma [Link et al., 1993, Correlation chart, Figure 5], although these ages are not known with certainty and have not been confirmed by isotopic dating methods; consequently, we consider the ages reported by Link et al. [1993] as preliminary.

‖Pole calculated after 10° correction to restore dike to a vertical orientation.

§Mean CWa pole has been calculated as described by Harlan [1993].

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nificantly north of those from sedimentary rocks of the Middle Proterozoic Belt Supergroup of western Montana and Sibley Group of Ontario (Figure 14a). The age of the Belt Supergroup rocks and age of their characteristic magnetization is poorly known and a matter of debate. Published estimates for the period of Belt deposition based on geochronologic and paleomagnetic data range from about 1500 to 900 Ma (see discussions by *Obradovich et al.* [1984], *Elston* [1984], *Winston* [1990], and *Link et al.* [1993]). Published paleomagnetic poles for the Belt [*Elston and Bressler*, 1980] are derived from the middle and upper parts of the Belt Supergroup; paleomagnetic results from the lower Belt have not yielded reliable data [*Elston and Bressler*, 1980; *Link et al.*, 1993]. Nominally, the Belt poles have been assigned ages of 1400 to 1250 Ma on the basis of arguments presented by *Link et al.* [1993]. However, none of them can be assigned unequivocal ages because the rocks from which they have been derived cannot be reliably dated by isotopic methods. Poles from the Middle Proterozoic Sibley Group [*Robertson*, 1973a] overlap those of most of the Belt Supergroup and that of the Laramie anorthosite complex/Sherman Granite due to the large uncertainty associated with both the S1 and S2 pole positions. The Sibley Group rocks have yielded a whole rock Rb-Sr isochron age of 1339 ± 33 Ma (recalculated from data in *Franklin* [1978], using the revised ^{87}Rb decay constant of *Steiger and Jäger* [1977]). The apparent discordance between poles with isotopic ages of 1480 to 1415 Ma with those of the Belt Supergroup and Sibley Group indicates that either igneous poles record significant northward translation or tilting relative to those of the sedimentary rocks or that the two groups of poles are of different age. Given that poles from igneous rocks from Wyoming to Labrador are in relatively good agreement, we consider significant tectonic displacement of the poles to be unlikely and prefer the interpretation that poles from the igneous and sedimentary rocks are of different ages. Tentatively, we suggest that the Belt poles must be younger than 1415 to 1460 Ma and that the apparent discordance between poles from igneous and sedimentary rocks is due to southward APW from about 1480 to 1350? Ma. The gross dissimilarity between Belt poles and those of the well-defined Keweenawan track indicates that, barring significant local vertical-axis rotation and assumed coherence of the Belt terrane with interior Laurentia, the Belt poles must be significantly older than 1100 Ma. Because the Belt and Sibley poles plot significantly southeast of the well-dated Coppermine volcanics/Mackenzie dikes (1267 Ma) and Sudbury dikes (1238 Ma), it is likely that the Belt Supergroup poles are considerably older than 1250 Ma. Solely on the basis of the paleomagnetic and geochronologic data shown in Figure 14a, we suggest that the age of the Belt Supergroup poles is about 1300 to 1400 Ma, slightly more restricted than the range proposed by *Elston and Bressler* [1980] and *Link et al.* [1993], but consistent with the Rb-Sr age from the Sibley Group rocks.

On the basis of published paleomagnetic poles listed in Table 5 and the paleomagnetic data presented in this study, we propose that the 1500 to 1350? Ma track of the Middle Proterozoic APW path is characterized by gross north to south movement, which corresponds to a component of southward translation of North America. The interval 1350? to 1250 Ma is characterized by overall northwestward APW from the Belt/Sibley poles to those of the Coppermine volcanics/Mackenzie dikes and Sudbury dikes, and this movement may reflect counterclockwise rotation of the craton. The direction and magnitude of APW between 1250 and 1110 Ma is unknown at present because of the lack of high-quality poles from this period of time [*Palmer and Halls*, 1986]. Tentatively, we have shown the path to track to high northerly paleolatitudes and connect with reverse polarity poles of the Midcontinent Rift. The extent to which these poles record apparent nondipole behavior of the geomagnetic field (i.e., reversal asymmetry) or high rates of APW has been controversial [*Nevanlinna and Pesonen*, 1983; *Pesonen and Halls*, 1983; *Lewchuk and Symons*, 1990; *Symons*, 1992; *Costanzo-Alvarez and Dunlop*, 1993; *Ernst and Buchan*, 1993]. The discovery of antiparallel reversals in both reverse and normal polarity chrons [*Symons*, 1989a; *Lewchuk and Symons*, 1990] recorded by rocks of the Midcontinent Rift has led some workers to conclude that the interval 1110 to 1090 Ma is marked by relatively rapid southward APW during this period of time. By analogy with Phanerozoic APW paths, relatively constant motion, such as that between the 1480 to 1415 Ma intrusions and the Belt/Sibley poles and 1110 to 1090 Ma Keweenawan track, may reflect relatively constant plate motions during these intervals. Sharp breaks in directions of APW such as that about 1350? Ma, or abrupt changes in paleolatitude such as at 1110 Ma, may record major changes in plate motion, although the causes remain speculative.

Age and Origin of Secondary Magnetization in Sherman Granite

The magnetization of southeast declination and shallow negative inclination observed at four sites in Sherman Granite (Figures 8c and 8d) is anomalous with respect to the characteristic magnetization observed in other Sherman Granite sites, the Laramie anorthosite complex, and with that of other circa 1400 Ma plutons elsewhere in North America. Similar magnetizations have been found in some sites in other 1400 Ma plutons elsewhere in the Front Range of Colorado (A. Lester and E. Larson, personal communication, 1991) and in at least one site in the 1070-1080 Ma Pikes Peak Granite located about 100 km to the south [*Harlan and Geissman*, 1989]. This direction is similar to a secondary magnetization reported from the 520 Ma McClure Mountain Complex exposed in the Wet Mountains of southern Colorado [*Lynnes and Van der Voo*, 1984] and to remagnetized Cambro-Ordovician dikes in the Pedernal uplift of New Mexico [*Jackson et al.*, 1988]. The origin of this magnetization is unknown but may be closely associated with the Precambrian/Paleozoic nonconformity and zones of penetrative fractures. The presence of this magnetization in rocks removed from the Precambrian/Paleozoic nonconformity throughout Colorado and New Mexico leads us to suggest that the remanence may be a thermochemical magnetization related to fluids associated with deformation and uplift of the Ancestral Rocky Mountains in Pennsylvanian and Permian time. A VGP calculated from the four sites (Figure 14b) is consistent with late Paleozoic poles from elsewhere in North America [*McCabe and Elmore*, 1989].

Conclusions

The Laramie anorthosite complex and Sherman Granite contain stable, well-defined magnetizations that were probably acquired during subsolidus cooling at about 1415 Ma. The ^{40}Ar/^{39}Ar dates indicate thermal remagnetization has not occurred. Both the thermochronologic and paleomagnetic data suggest that the poles average secular variation and are reliable recorders of the geomagnetic field. Paleomagnetic pole positions derived from both plutons are indistinguishable at the 95% confidence level and are combined to yield a mean pole at 215.0°E, 6.7°S (A_{95} = 3.5°). Although uncertainties exist regarding possible unrecognized tilts, this pole is consistent with poles derived from rocks of similar age elsewhere in North America. Locally, near the Precambrian/Paleozoic unconformity or near zones of penetrative fractures, the Sherman Granite has been partially to completely remagnetized during late Paleozoic time, possibly as a consequence of fluid flow associated with uplift of the Ancestral Rocky Mountains.

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S.S. Harlan and L.W. Snee, U.S. Geological Survey, P.O. Box 25046, MS 963, Federal Center, Denver, CO 80225 (e-mail: sharlan@greenwood.cr.usgs.gov)

A.J. Brearley and J.W. Geissman, Department of Earth and Planetary Sciences, University of New Mexico, Albuquerque, NM 87131.

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